26016 Concourse viewing platform
Model v19m - ULS member force envelopes (SOFiSTiK WinGRAF)

Prepared 13 Nov 2026 from 26016_check_v19m (v19 + Mz release at the 18 Level 1 r 8.5 ring ends), read from a copy of the database.

Envelope: the model's own MAXIMA superposition COMB 3 (EXTR NONL, TYPE DESI, BASE 4300): each of the 43 ULS STR combinations 101-130, 135, 141-148, 151-154 enters as an exclusive alternative (TYPE AG1, factor 1.00), so every envelope value is the value of one complete combination. EQU combinations 131-134 are not included. The combinations are linear (ASE SYST PROB LINE) COPY combinations.

Beam envelope load cases: N 4321/4322, Vy 4323/4324, Vz 4325/4326, Mt 4327/4328, My 4329/4330, Mz 4331/4332 (max/min). SLS characteristic 201-221: 4421-4432 (same order).

Check: at every result station of every beam the MAXIMA value equals the largest / smallest value read from the 43 combinations (largest difference over all stations: N 0.0e+00, My 2.4e-09 kNm).

Sign convention: SOFiSTiK local beam axes, N > 0 tension, My about the local y (strong) axis, Mz about the local z (weak) axis. Red = max load case, blue = min load case. Units kN, kNm.

Floor framing is drawn in a steep isometric view because My, Vz act in the vertical plane and cannot be read in a true top view (tested: diagrams rotated into the plan flip side with the member direction). Mz of Level 1 is drawn in a true plan view (X right, Y up).

Kingpost / central mast: modelled with shell elements (QUAD), no beam group, so no beam N / My diagram is given here.

ULS envelope per member group - forces N, Vy, Vz

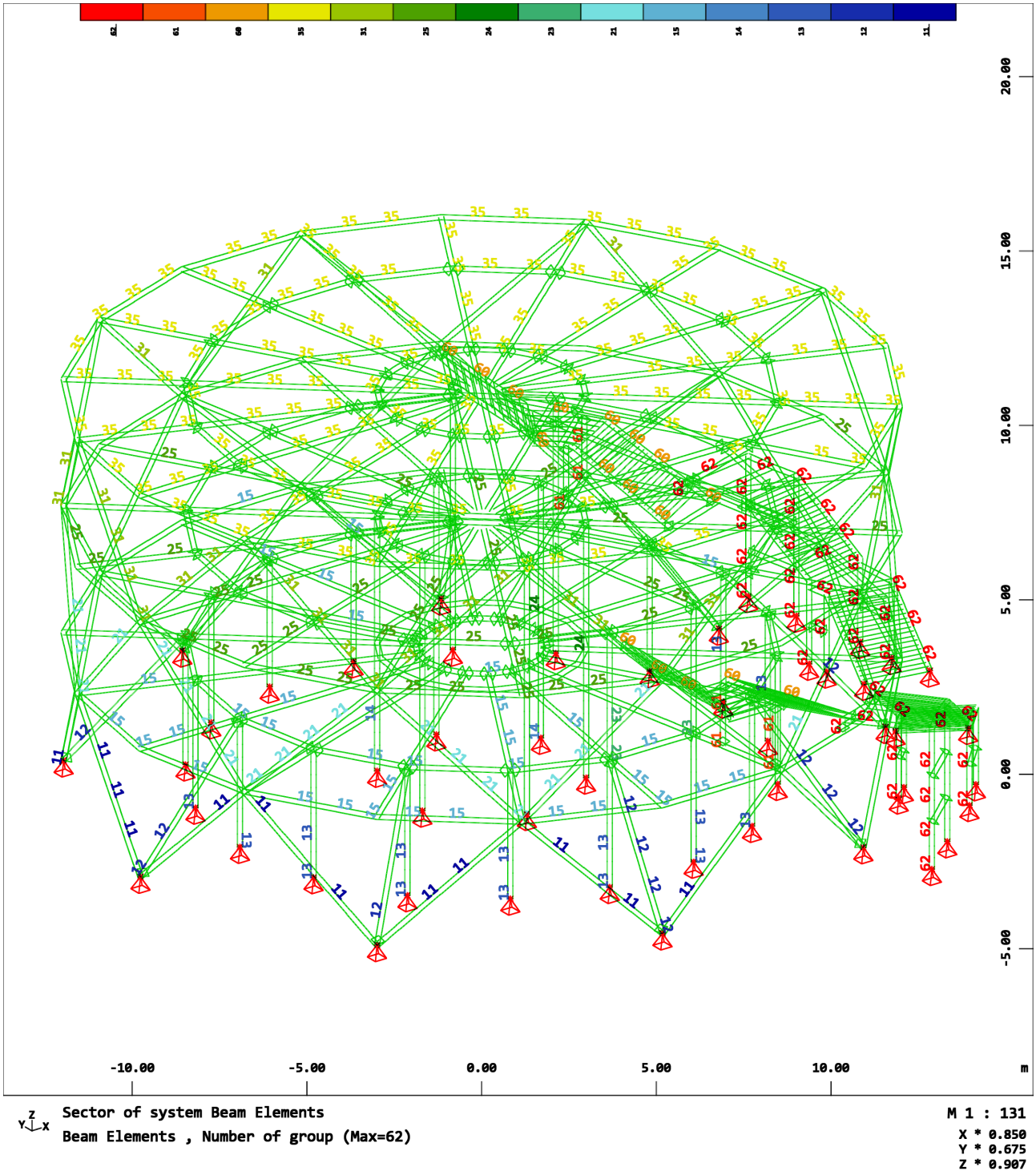
Each cell: envelope value, governing combination (LC), beam element and x [m from beam start]. Full data incl. SLN: env_table_ULS.csv / .json.

Group							
11	V-column rakers, ground to L1 SHS 150x150x10	55.7 (LC 109) b 60197 x 0.00	-294.0 (LC 121) b 60239 x 0.00	1.6 (LC 124) b 60141 x 0.00	-1.8 (LC 126) b 60084 x 0.42	2.5 (LC 120) b 60071 x 0.00	-2.5 (LC 106) b 60154 x 0.42
12	rakers, ground to L1 SHS 150x150x10	56.8 (LC 116) b 70104 x 0.41	-84.1 (LC 113) b 70053 x 0.00	1.3 (LC 130) b 70079 x 0.00	-1.6 (LC 125) b 70092 x 0.00	1.5 (LC 115) b 70001 x 0.00	-1.5 (LC 123) b 70013 x 0.41
13	vertical columns (ground-L1, L1-L2) SHS 150x150x10	26.5 (LC 110) b 80061 x 0.00	-267.8 (LC 121) b 80050 x 0.41	9.0 (LC 124) b 80011 x 0.00	-6.3 (LC 124) b 80030 x 0.41	7.8 (LC 127) b 80141 x 0.00	-7.7 (LC 128) b 80151 x 0.00
14	vertical columns, ground to L1 SHS 150x150x10	-7.9 (LC 109) b 90011 x 0.00	-399.0 (LC 135) b 90030 x 0.41	0.8 (LC 152) b 90051 x 0.00	-1.4 (LC 151) b 90011 x 0.00	0.3 (LC 113) b 90010 x 0.41	-0.3 (LC 112) b 90040 x 0.41
15	Level 1 framing (radials + rings) UB 254x146x43	109.3 (LC 153) b 100507 x 0.00	-71.6 (LC 112) b 100832 x 0.39	50.1 (LC 122) b 100841 x 0.25	-50.1 (LC 120) b 100859 x 0.25	77.0 (LC 153) b 100842 x 0.00	-76.7 (LC 153) b 100868 x 0.39
21	V-column rakers, L1 to L2 SHS 150x150x10	59.7 (LC 109) b 110085 x 0.00	-228.0 (LC 145) b 110238 x 0.41	4.6 (LC 120) b 110239 x 0.00	-4.5 (LC 120) b 110014 x 0.41	7.1 (LC 122) b 110127 x 0.00	-6.8 (LC 120) b 110182 x 0.41
23	vertical columns, L1 to L2 SHS 150x150x10	-3.4 (LC 109) b 120001 x 0.00	-144.7 (LC 121) b 120020 x 0.40	3.6 (LC 154) b 120001 x 0.00	-4.1 (LC 152) b 120020 x 0.40	0.8 (LC 128) b 120010 x 0.40	-0.6 (LC 123) b 120020 x 0.40
24	vertical columns, L1 to L2 SHS 150x150x10	1.8 (LC 109) b 130021 x 0.00	-264.0 (LC 146) b 130040 x 0.40	7.2 (LC 109) b 130051 x 0.00	-10.6 (LC 125) b 130060 x 0.40	8.0 (LC 110) b 130040 x 0.40	-8.4 (LC 124) b 130060 x 0.40
25	Level 2 framing (radials + rings) UB 254x146x43; UB 457x191x89	62.3 (LC 145) b 140632 x 0.00	-55.4 (LC 110) b 140748 x 0.00	27.2 (LC 119) b 140775 x 0.60	-26.2 (LC 119) b 140439 x 0.00	146.2 (LC 146) b 140684 x 0.00	-109.4 (LC 145) b 140389 x 0.39
31	V-column rakers, L2 to roof SHS 150x150x10	66.8 (LC 112) b 150085 x 0.00	-60.0 (LC 147) b 150099 x 0.00	5.9 (LC 108) b 150070 x 0.41	-5.9 (LC 110) b 150127 x 0.00	7.2 (LC 153) b 150127 x 0.00	-7.1 (LC 106) b 150112 x 0.41
35	roof framing (radials + rings) UB 254x146x43	72.8 (LC 113) b 160070 x 0.42	-67.4 (LC 112) b 160030 x 0.42	1.6 (LC 125) b 160129 x 0.42	-1.4 (LC 126) b 160070 x 0.42	31.0 (LC 110) b 160627 x 0.39	-31.3 (LC 108) b 160081 x 0.00
60	internal stair (stringers, landing, treads) RHS 200x100x6; SHS 50x50x5; H = 5 mm	57.8 (LC 120) b 170329 x 0.34	-22.1 (LC 135) b 170316 x 0.00	22.5 (LC 122) b 170330 x 0.30	-27.8 (LC 152) b 170138 x 0.00	37.5 (LC 122) b 170376 x 0.00	-28.5 (LC 152) b 170008 x 0.00
61	internal stair landing posts SHS 100x100x5	3.8 (LC 107) b 180001 x 0.00	-51.6 (LC 120) b 180010 x 0.40	0.0 (LC 125) b 180009 x 0.00	-0.0 (LC 135) b 180002 x 0.00	0.0 (LC 135) b 180012 x 0.00	-0.0 (LC 107) b 180012 x 0.00
62	external stair ground to L1 RHS 200x100x6; H = 5 mm; SHS 100x100x5; SHS 80x80x6	65.0 (LC 111) b 190321 x 0.05	-62.7 (LC 135) b 190458 x 0.41	27.2 (LC 135) b 190016 x 0.00	-28.8 (LC 135) b 190161 x 0.00	27.9 (LC 121) b 190159 x 0.00	-15.6 (LC 135) b 190388 x 0.24

ULS envelope per member group - moments Mt, My, Mz

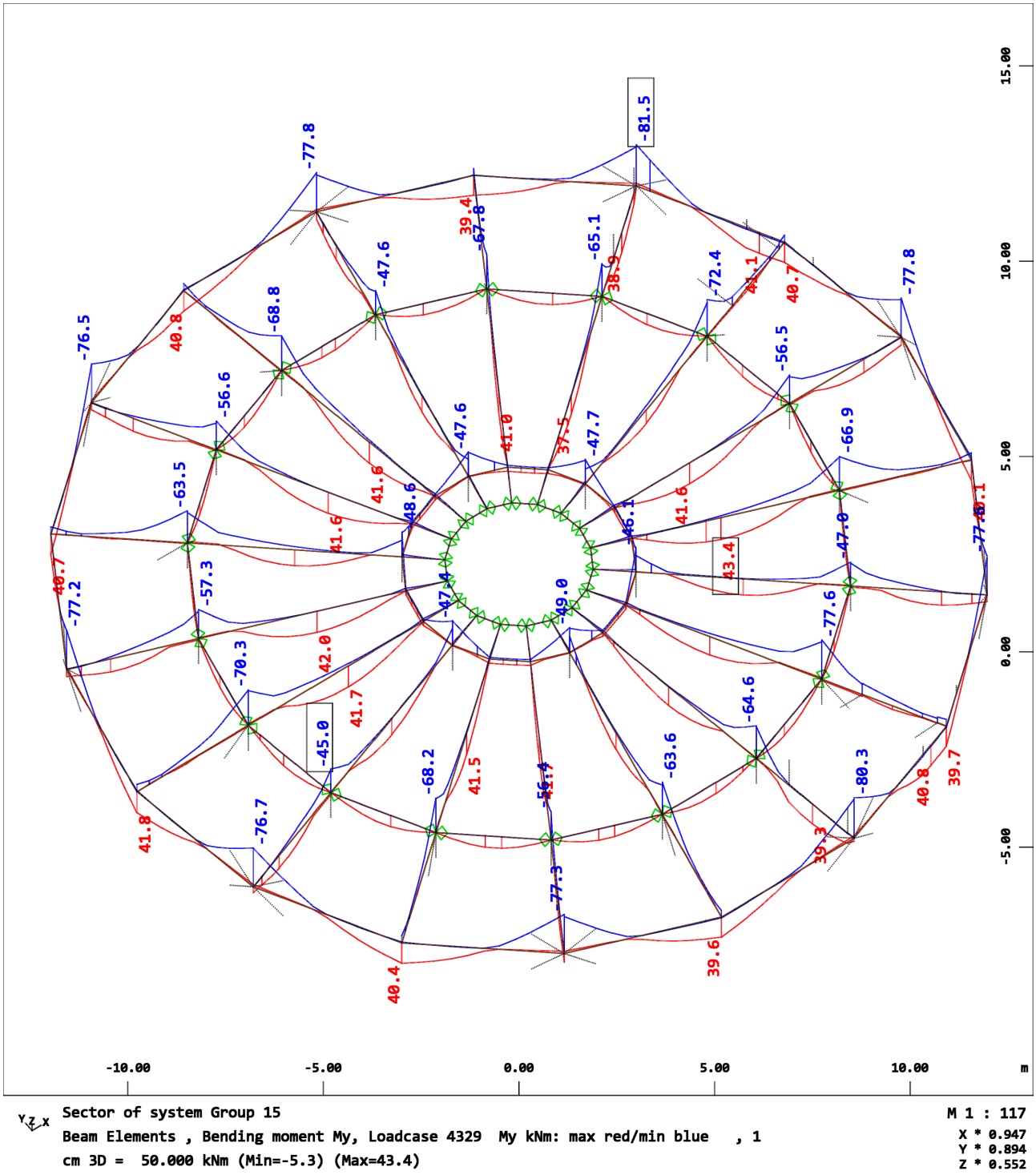
Each cell: envelope value, governing combination (LC), beam element and x [m from beam start]. Full data incl. SLN: env_table_ULS.csv / .json.

Group							
11	V-column rakers, ground to L1 SHS 150x150x10	1.1 (LC 152) b 60167 x 0.00	-1.1 (LC 152) b 60058 x 0.00	3.1 (LC 154) b 60141 x 0.00	-4.7 (LC 151) b 60113 x 0.00	5.8 (LC 152) b 60084 x 0.42	-2.1 (LC 121) b 60154 x 0.42
12	rakers, ground to L1 SHS 150x150x10	0.5 (LC 130) b 70010 x 0.00	-0.6 (LC 125) b 70010 x 0.00	2.0 (LC 115) b 70007 x 0.41	-0.0 (LC 118) b 70078 x 0.41	3.0 (LC 125) b 70104 x 0.41	-2.2 (LC 129) b 70013 x 0.00
13	vertical columns (ground-L1, L1-L2) SHS 150x150x10	0.2 (LC 112) b 80183 x 0.00	-0.6 (LC 152) b 80200 x 0.00	7.1 (LC 118) b 80015 x 0.41	-7.1 (LC 125) b 80015 x 0.41	9.7 (LC 152) b 80200 x 0.40	-10.9 (LC 152) b 80021 x 0.00
14	vertical columns, ground to L1 SHS 150x150x10	0.0 (LC 151) b 90005 x 0.00	-0.0 (LC 151) b 90023 x 0.00	1.0 (LC 112) b 90031 x 0.00	-1.3 (LC 113) b 90001 x 0.00	3.1 (LC 152) b 90051 x 0.00	-5.5 (LC 151) b 90011 x 0.00
15	Level 1 framing (radials + rings) UB 254x146x43	1.2 (LC 135) b 100859 x 0.00	-1.1 (LC 135) b 100841 x 0.00	43.4 (LC 151) b 100436 x 0.39	-81.5 (LC 152) b 100832 x 0.39	10.1 (LC 120) b 100841 x 0.00	-10.4 (LC 120) b 100859 x 0.00
21	V-column rakers, L1 to L2 SHS 150x150x10	2.2 (LC 120) b 110011 x 0.00	-2.3 (LC 121) b 110251 x 0.00	21.9 (LC 121) b 110140 x 0.41	-22.6 (LC 120) b 110197 x 0.00	9.4 (LC 121) b 110183 x 0.00	-16.0 (LC 119) b 110252 x 0.41
23	vertical columns, L1 to L2 SHS 150x150x10	0.7 (LC 103) b 120015 x 0.00	-0.2 (LC 125) b 120001 x 0.00	3.4 (LC 128) b 120010 x 0.40	-2.5 (LC 123) b 120020 x 0.40	10.2 (LC 152) b 120020 x 0.40	-6.4 (LC 151) b 120020 x 0.40
24	vertical columns, L1 to L2 SHS 150x150x10	0.2 (LC 109) b 130041 x 0.00	-0.2 (LC 111) b 130041 x 0.00	6.0 (LC 130) b 130040 x 0.40	-6.0 (LC 123) b 130030 x 0.40	10.9 (LC 121) b 130060 x 0.40	-13.4 (LC 153) b 130051 x 0.00
25	Level 2 framing (radials + rings) UB 254x146x43; UB 457x191x89	1.0 (LC 120) b 140490 x 0.00	-1.2 (LC 148) b 140489 x 0.00	233.7 (LC 120) b 140698 x 0.39	-363.8 (LC 145) b 140389 x 0.39	11.2 (LC 119) b 140445 x 0.00	-9.3 (LC 121) b 140775 x 0.60
31	V-column rakers, L2 to roof SHS 150x150x10	5.0 (LC 111) b 150194 x 0.00	-5.0 (LC 111) b 150172 x 0.00	21.2 (LC 104) b 150070 x 0.41	-23.6 (LC 153) b 150127 x 0.00	20.1 (LC 108) b 150057 x 0.00	-15.0 (LC 145) b 150169 x 0.00
35	roof framing (radials + rings) UB 254x146x43	0.0 (LC 112) b 160815 x 0.00	-0.1 (LC 108) b 160782 x 0.00	74.0 (LC 107) b 160735 x 0.39	-60.8 (LC 107) b 160388 x 0.00	1.8 (LC 126) b 160070 x 0.42	-1.9 (LC 125) b 160119 x 0.42
60	internal stair (stringers, landing, treads) RHS 200x100x6; SHS 50x50x5; H = 5 mm	5.0 (LC 135) b 170136 x 0.00	-4.7 (LC 135) b 170377 x 0.32	16.7 (LC 122) b 170403 x 0.30	-22.6 (LC 120) b 170305 x 0.16	7.3 (LC 120) b 170328 x 0.00	-12.4 (LC 122) b 170137 x 0.00
61	internal stair landing posts SHS 100x100x5	0.2 (LC 121) b 180006 x 0.00	-0.2 (LC 122) b 180016 x 0.00	0.0 (LC 112) b 180004 x 0.40	-0.0 (LC 121) b 180007 x 0.00	0.0 (LC 126) b 180004 x 0.40	-0.0 (LC 152) b 180014 x 0.40
62	external stair ground to L1 RHS 200x100x6; H = 5 mm; SHS 100x100x5; SHS 80x80x6	7.4 (LC 135) b 190173 x 0.00	-7.2 (LC 135) b 190028 x 0.00	20.9 (LC 121) b 190161 x 0.00	-24.8 (LC 121) b 190159 x 0.00	18.2 (LC 121) b 190161 x 0.00	-24.3 (LC 122) b 190016 x 0.00



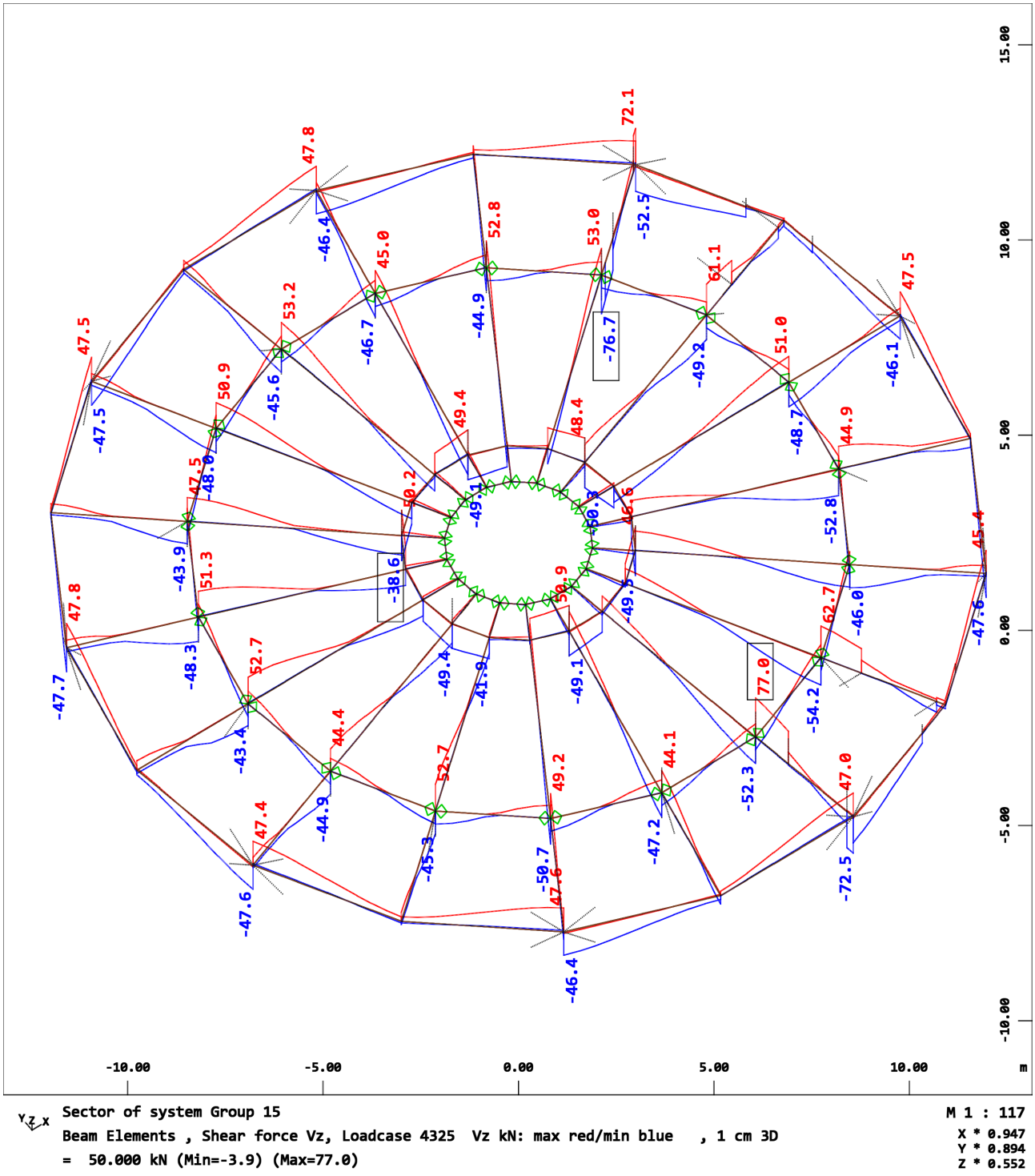
P00_key_beam_groups

Key: beam groups of model v19m (group number on each member, colour legend at top). Groups 1, 2, 6, 7 are shell (QUAD) groups and are not shown.



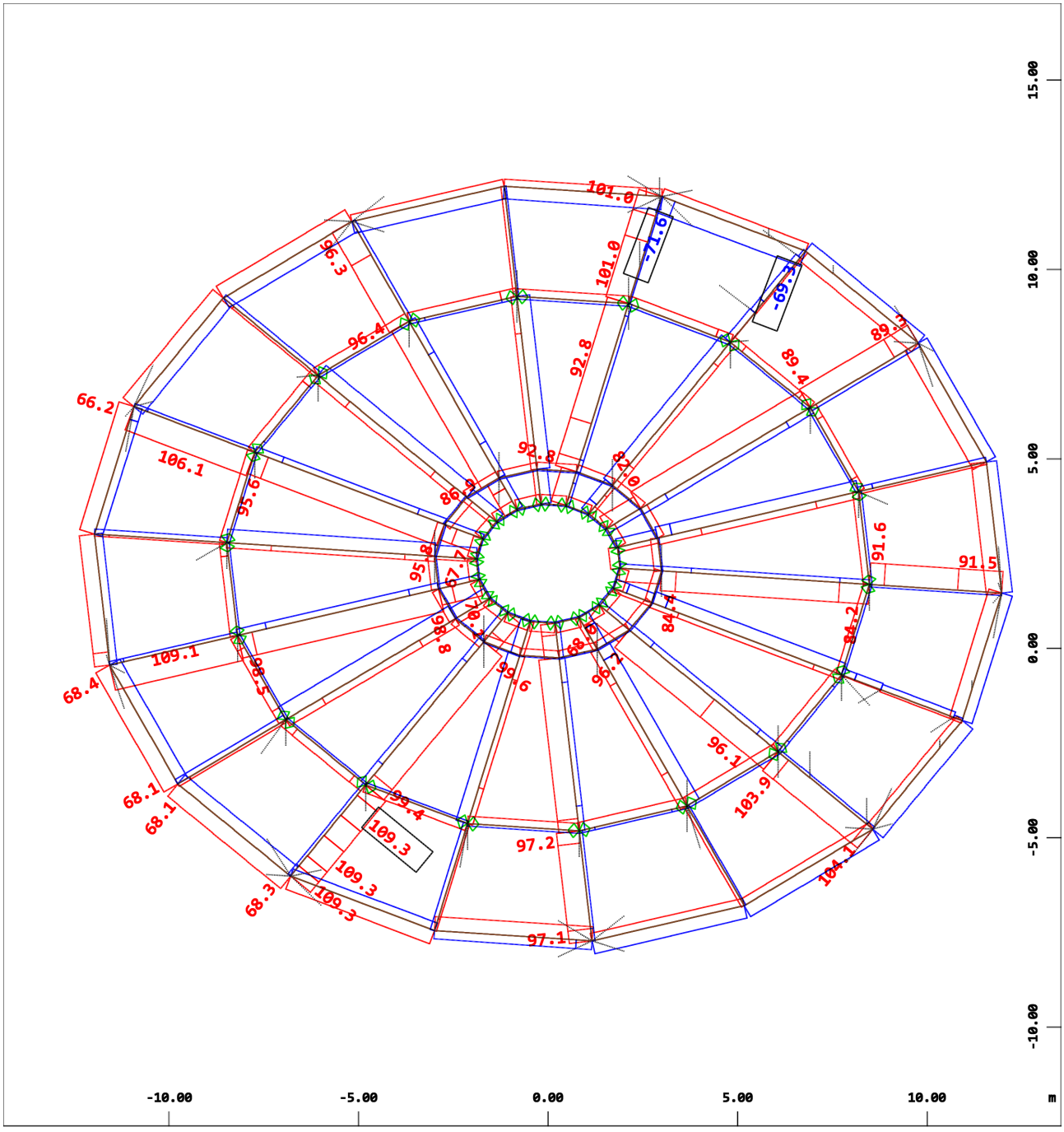
P01_L1_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 framing (group 15) My. Diagram scale 1 cm = 50 kNm (on the 25.5 cm sheet); labels only where |value| > 37 kNm. Envelope extremes: max 43.35, min -81.45 kNm.



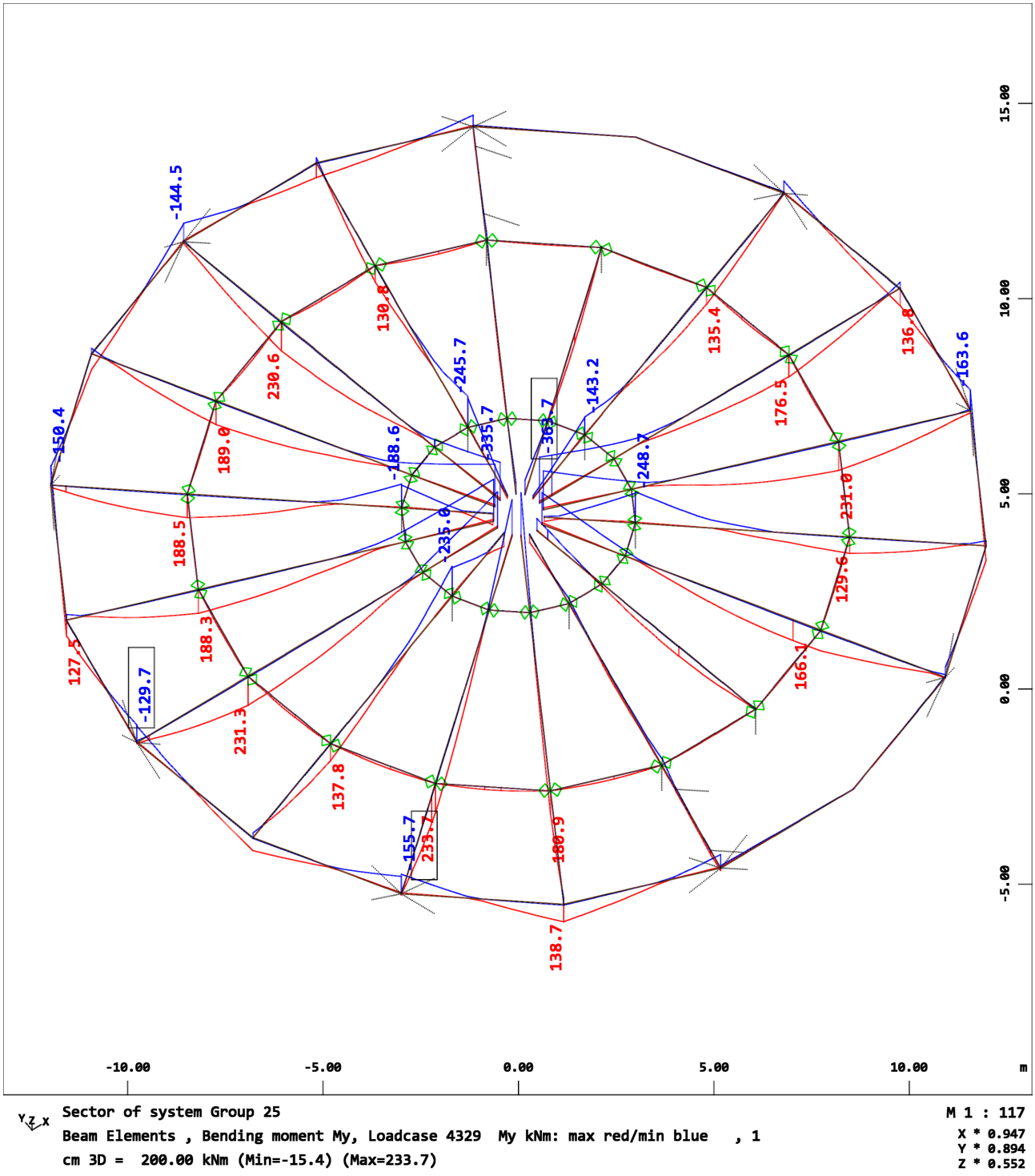
P02_L1_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 framing (group 15) Vz. Diagram scale 1 cm = 50 kN (on the 25.5 cm sheet); labels only where |value| > 35 kN. Envelope extremes: max 76.97, min -76.7 kN.



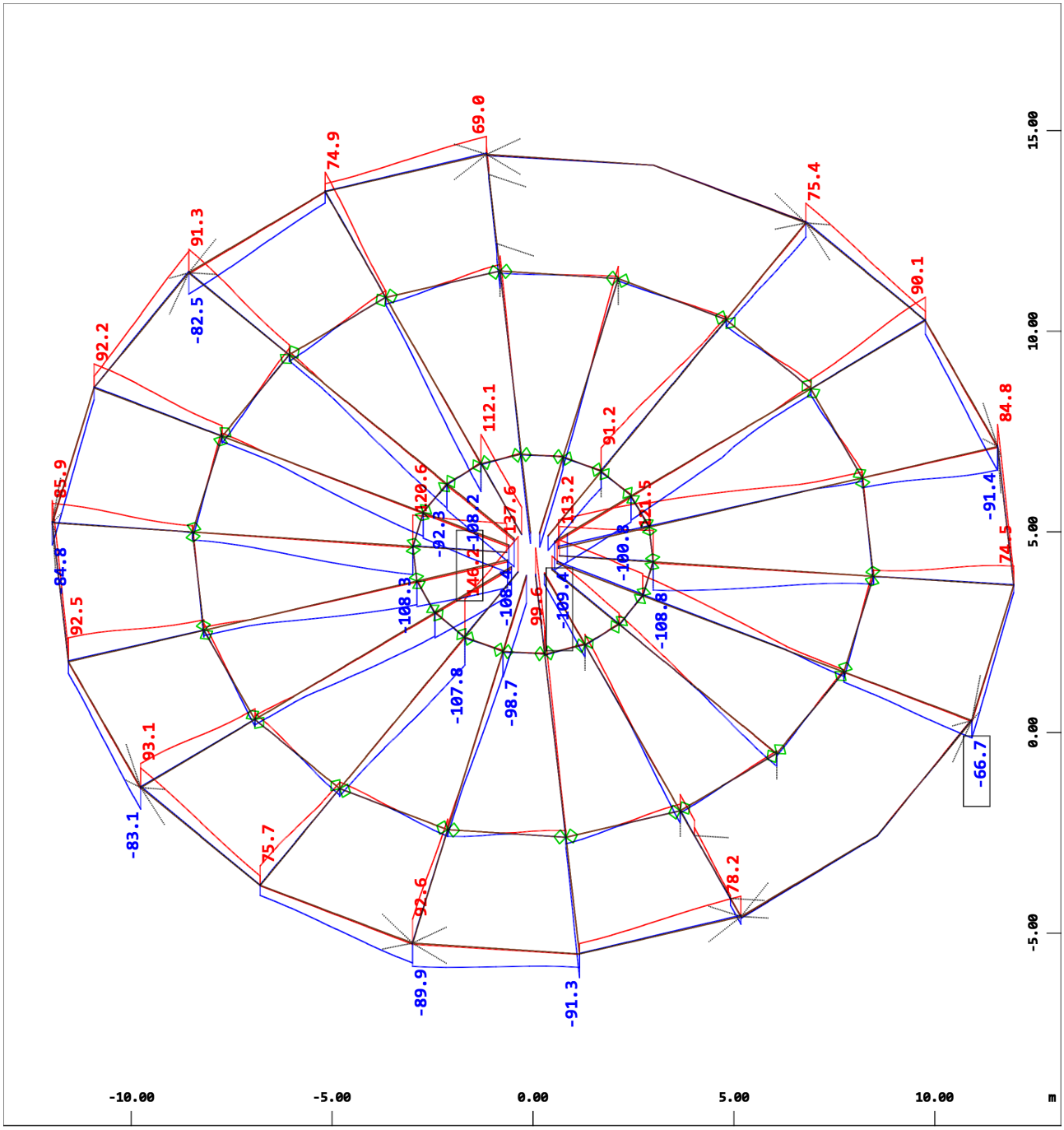
P03_L1_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 framing (group 15) N. Diagram scale 1 cm = 200 kN (on the 25.5 cm sheet); labels only where |value| > 66 kN. Envelope extremes: max 109.33, min -71.62 kN.



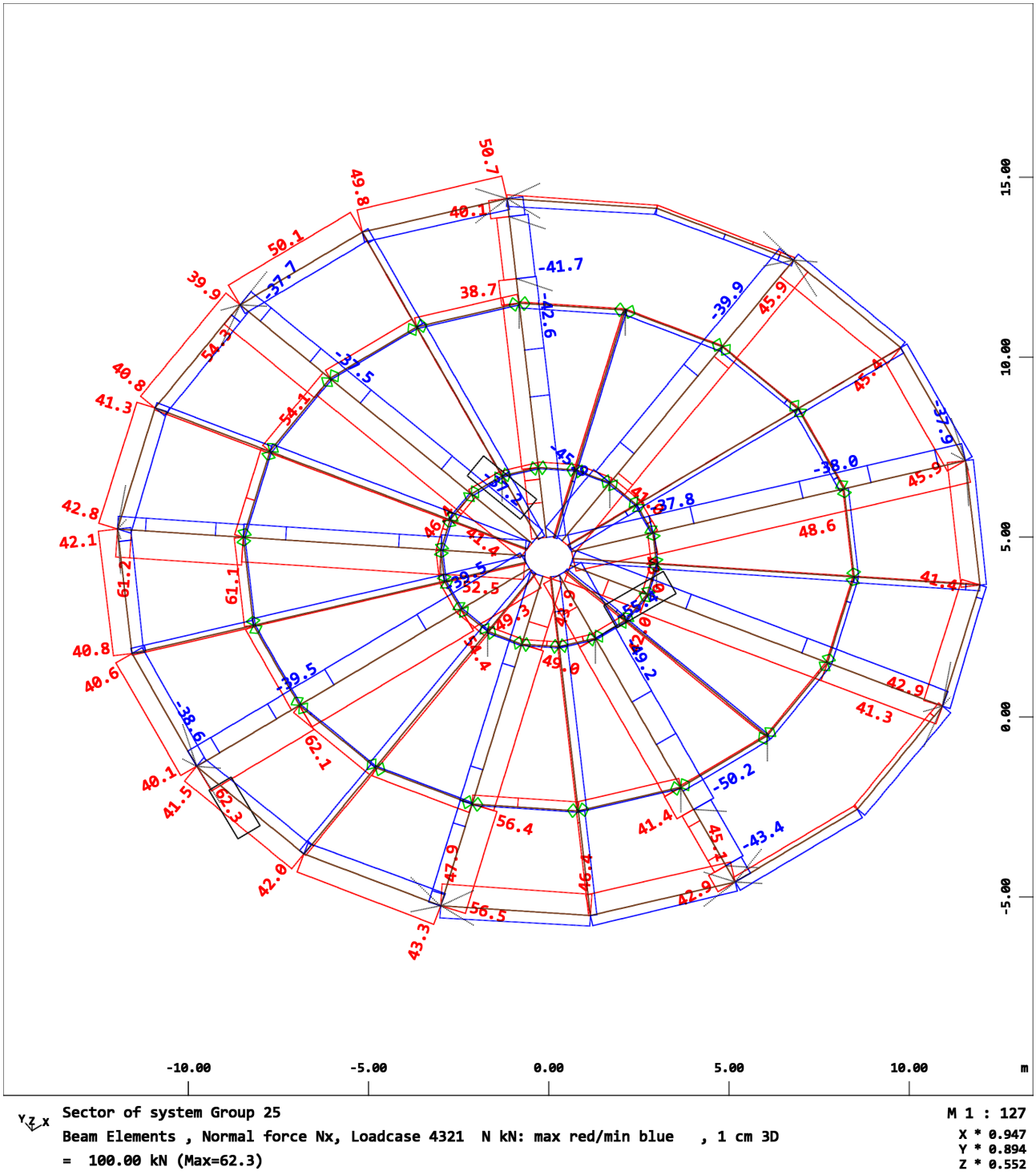
P04_L2_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 2 framing (group 25) My. Diagram scale 1 cm = 200 kNm (on the 25.5 cm sheet); labels only where |value| > 127 kNm. Envelope extremes: max 233.68, min -363.75 kNm.



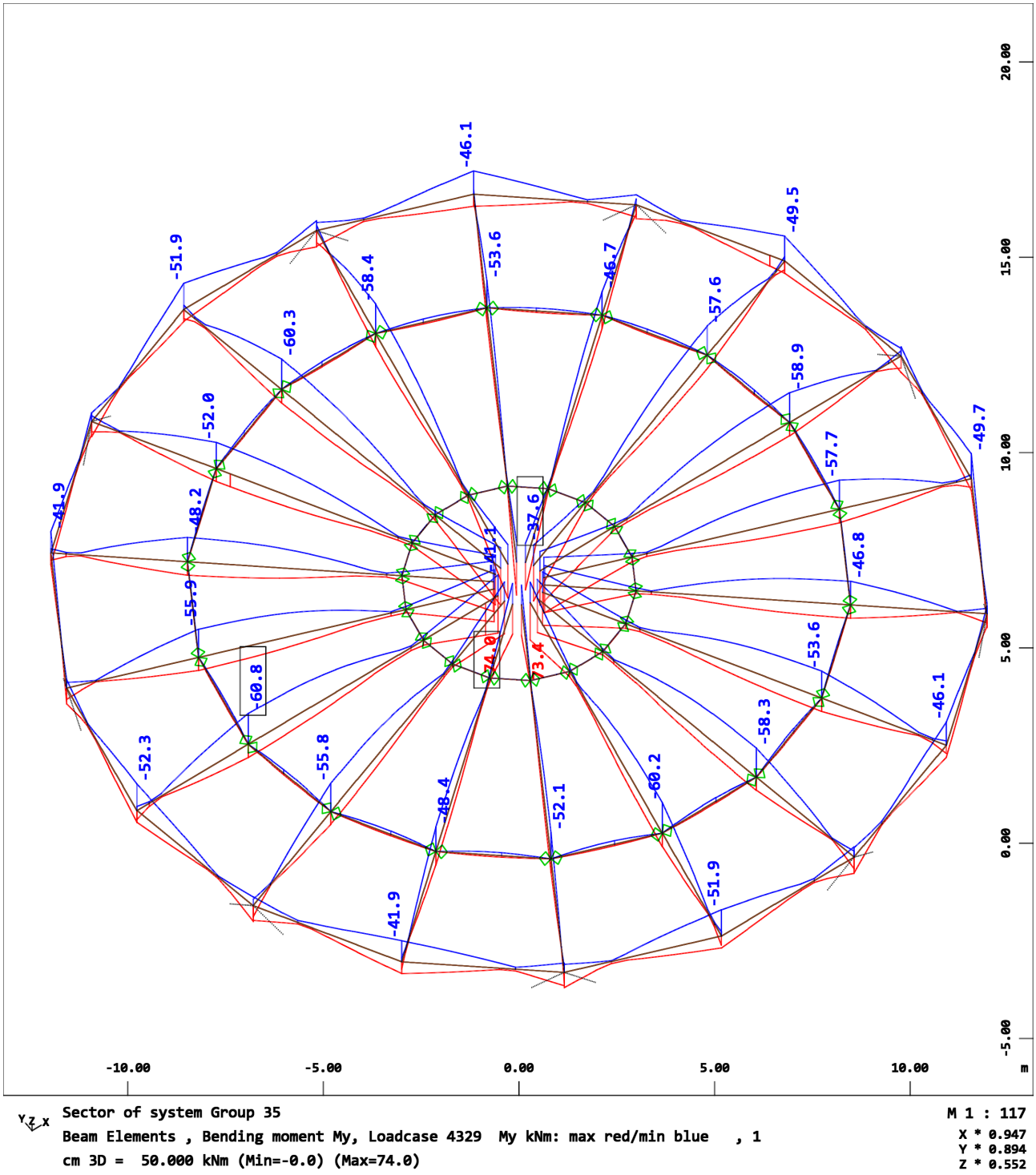
P05_L2_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 2 framing (group 25) Vz. Diagram scale 1 cm = 100 kN (on the 25.5 cm sheet); labels only where |value| > 66 kN. Envelope extremes: max 146.23, min -109.44 kN.



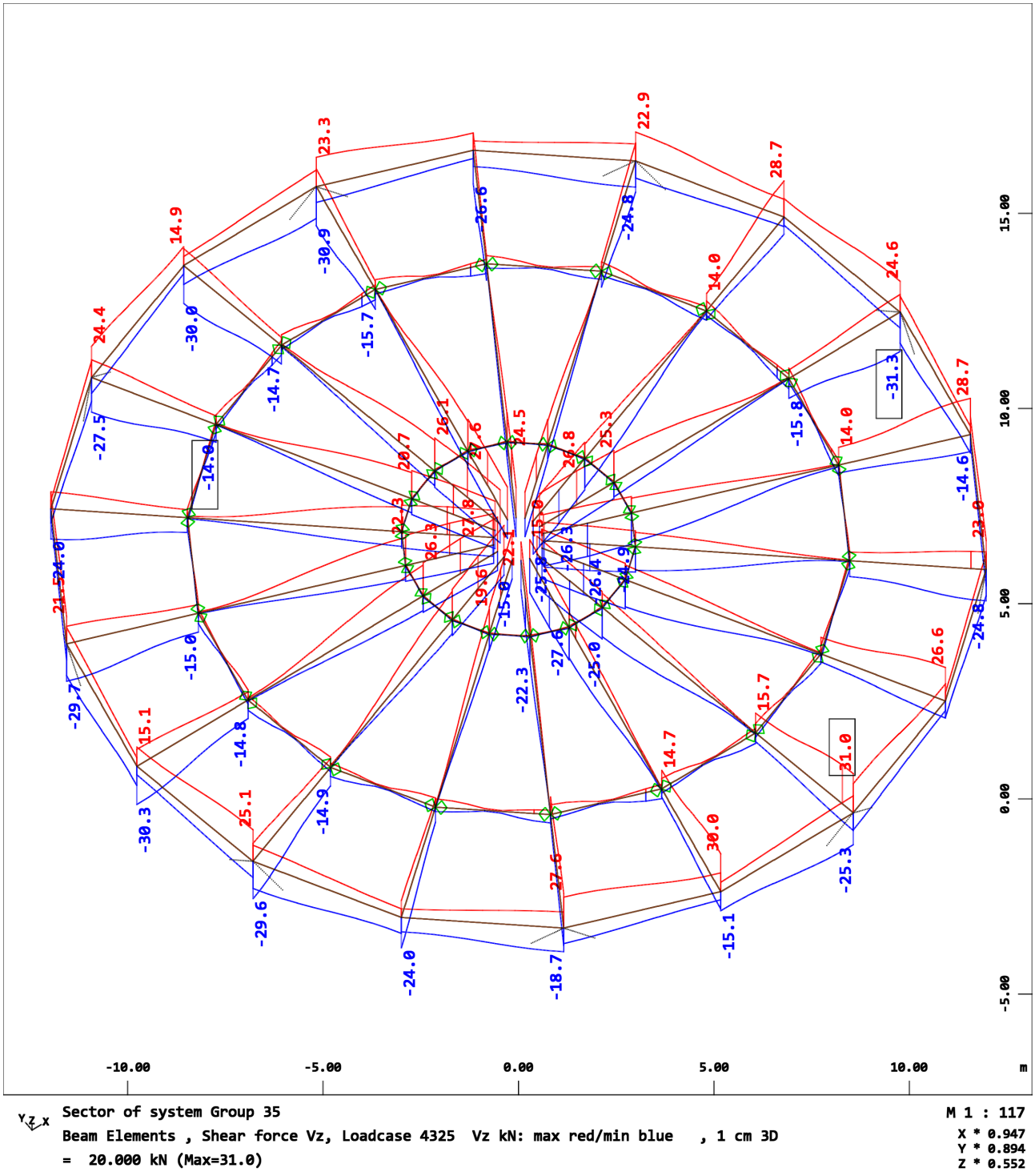
P06_L2_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 2 framing (group 25) N. Diagram scale 1 cm = 100 kN (on the 25.5 cm sheet); labels only where |value| > 37 kN. Envelope extremes: max 62.26, min -55.38 kN.



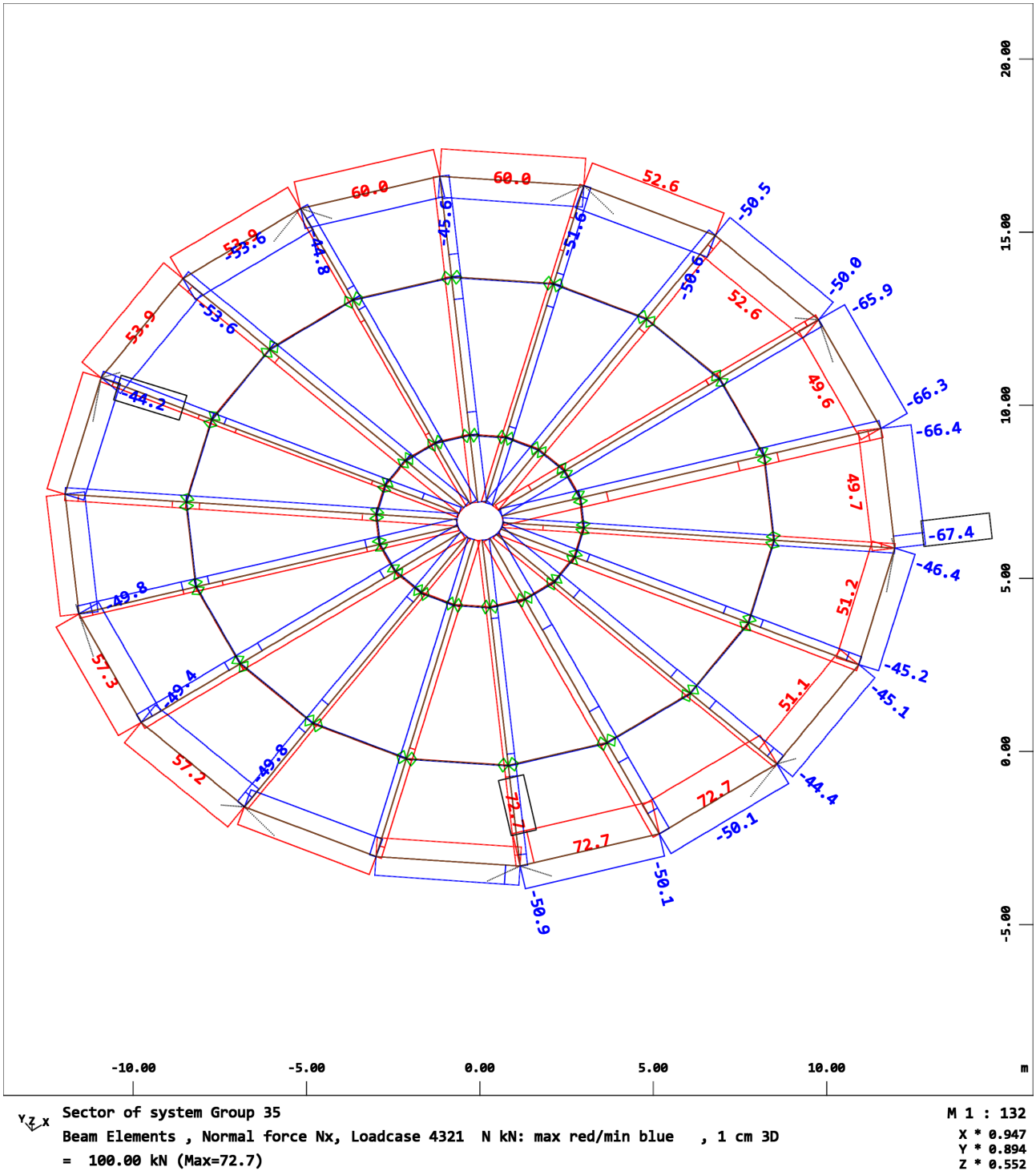
P07_RF_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Roof framing (group 35) My. Diagram scale 1 cm = 50 kNm (on the 25.5 cm sheet); labels only where |value| > 33 kNm. Envelope extremes: max 73.98, min -60.84 kNm.



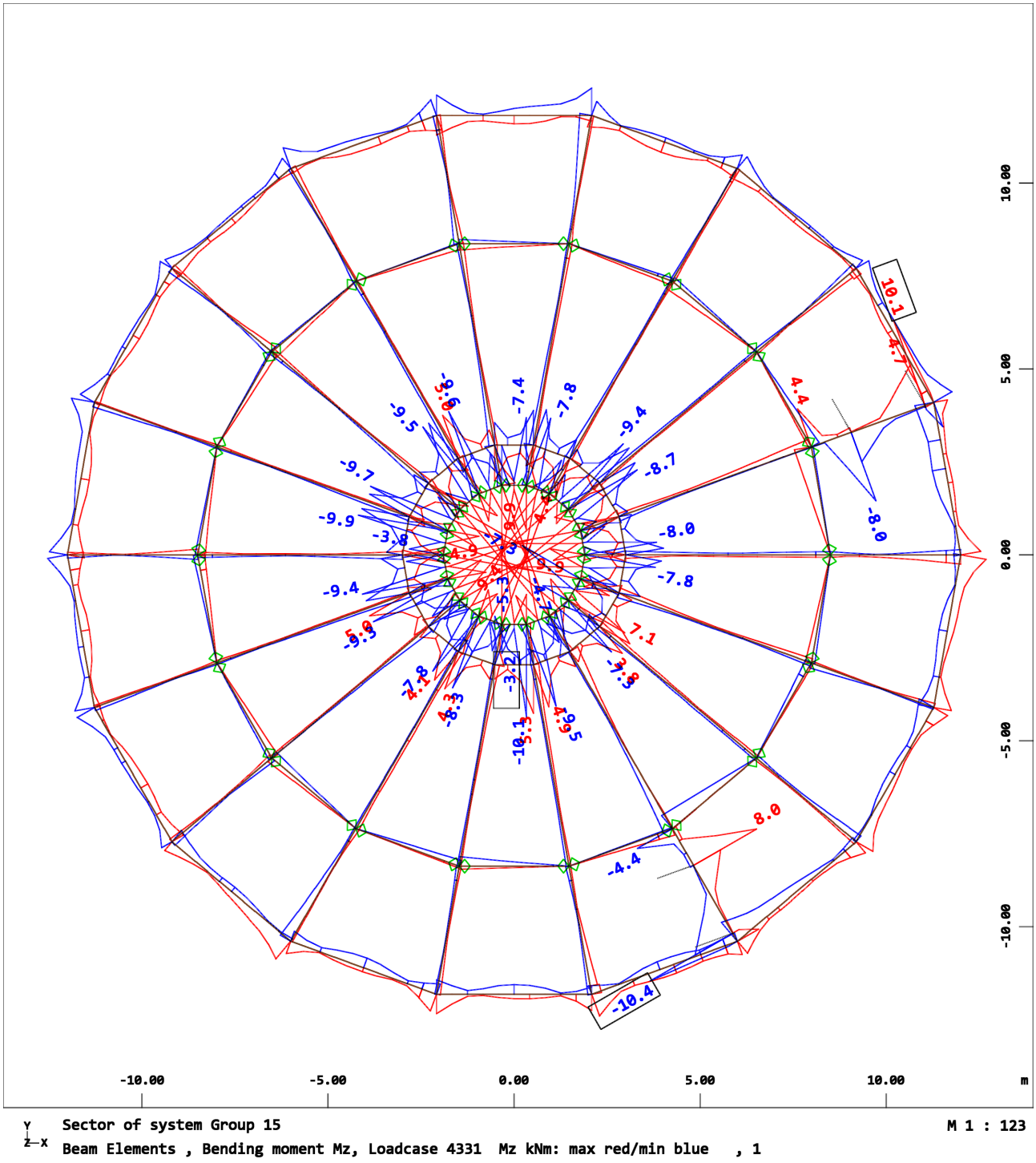
P08_RF_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Roof framing (group 35) Vz. Diagram scale 1 cm = 20 kN (on the 25.5 cm sheet); labels only where |value| > 14 kN. Envelope extremes: max 30.99, min -31.3 kN.



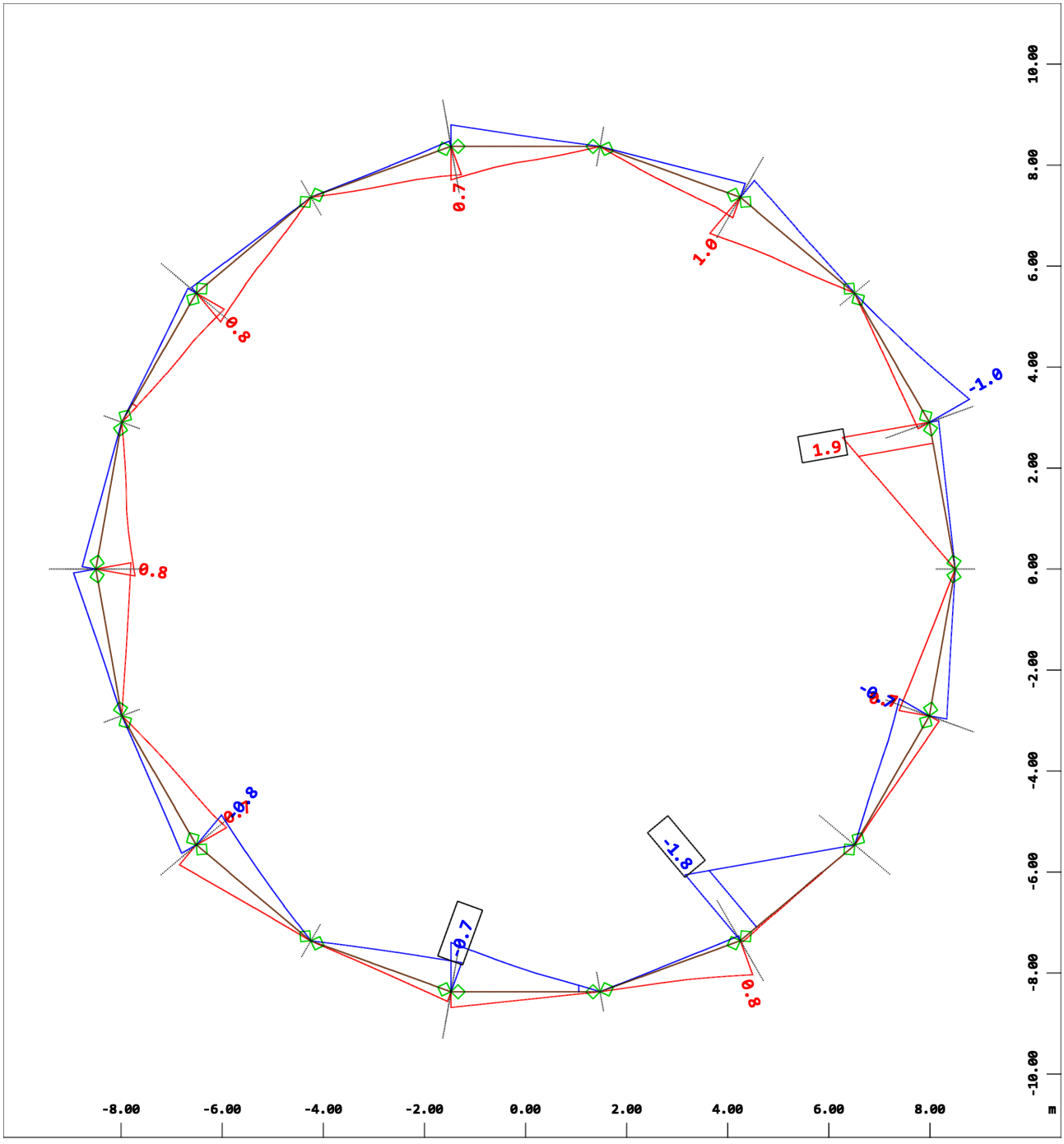
P09_RF_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Roof framing (group 35) N. Diagram scale 1 cm = 100 kN (on the 25.5 cm sheet); labels only where |value| > 44 kN. Envelope extremes: max 72.75, min -67.43 kN.



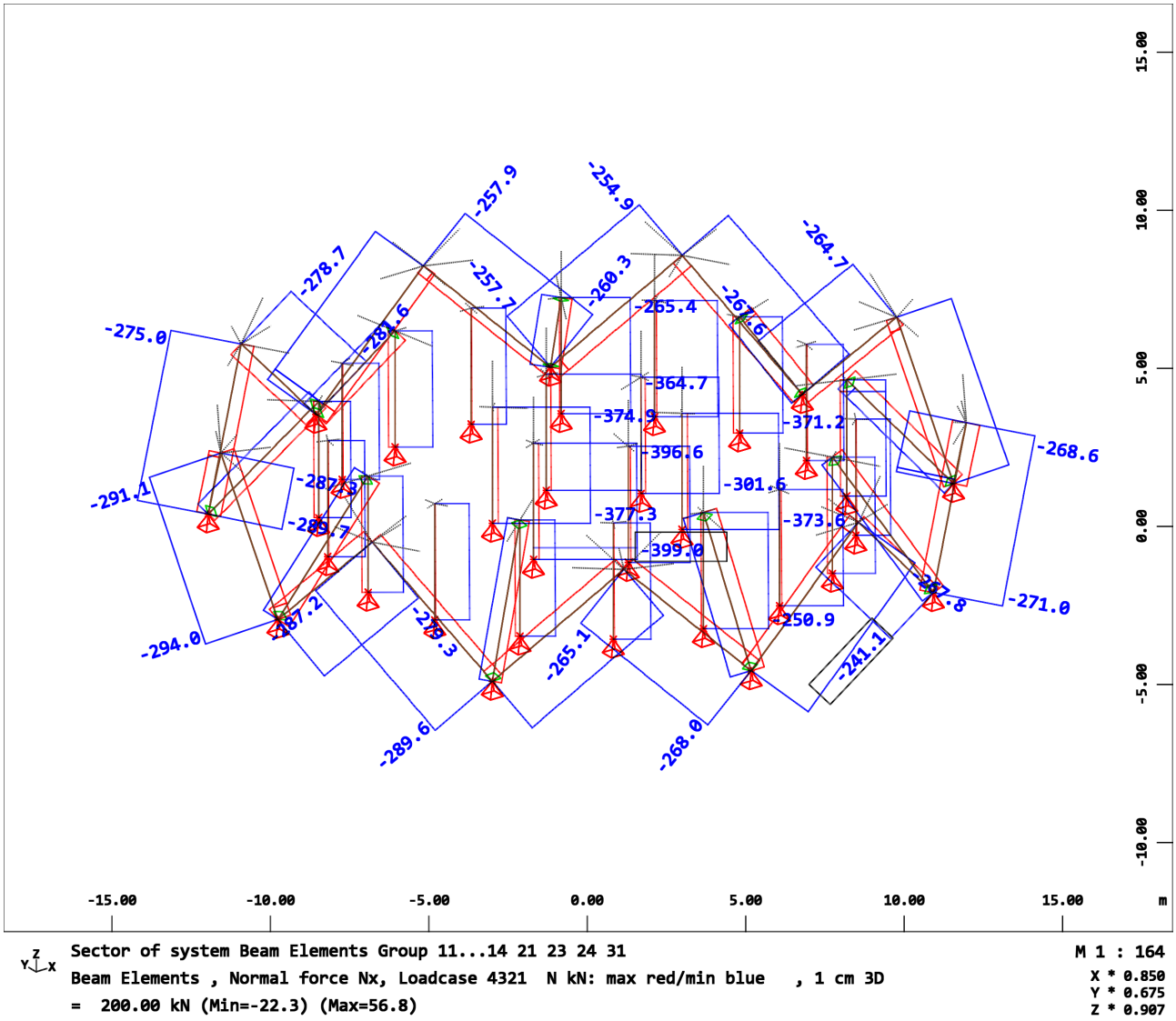
P10_L1_Mz_plan

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 framing (group 15) Mz, plan. Diagram scale 1 cm = 5 kNm (on the 25.5 cm sheet); labels only where |value| > 3.1 kNm. Envelope extremes: max 10.07, min -10.41 kNm.



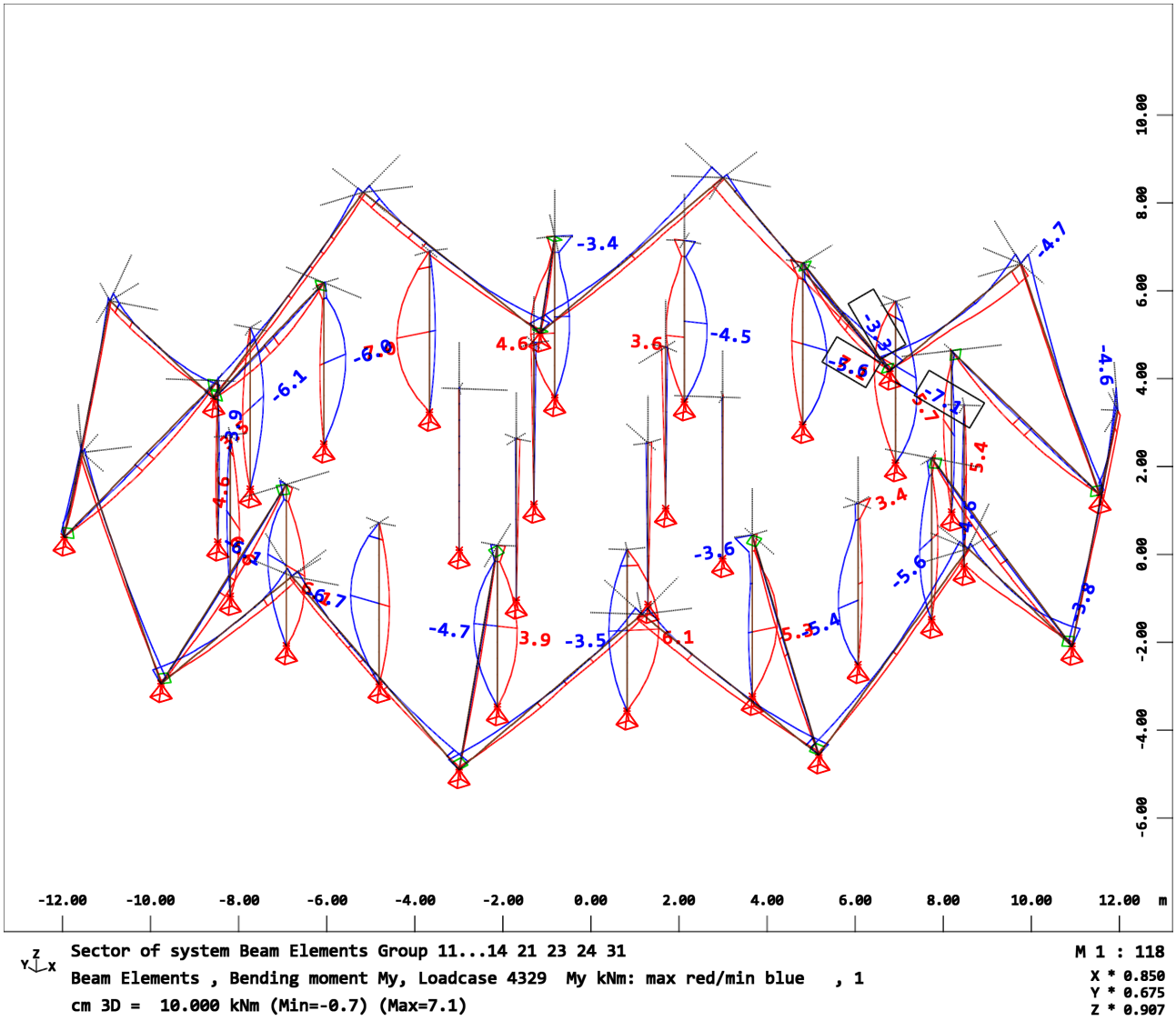
P11_L1_ring85_Mz_plan

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. Level 1 ring r 8.5 only (group 15) Mz, plan. Diagram scale 1 cm = 1 kNm (on the 25.5 cm sheet); labels only where |value| > 0.6 kNm. Envelope extremes: max 1.87, min -1.83 kNm.



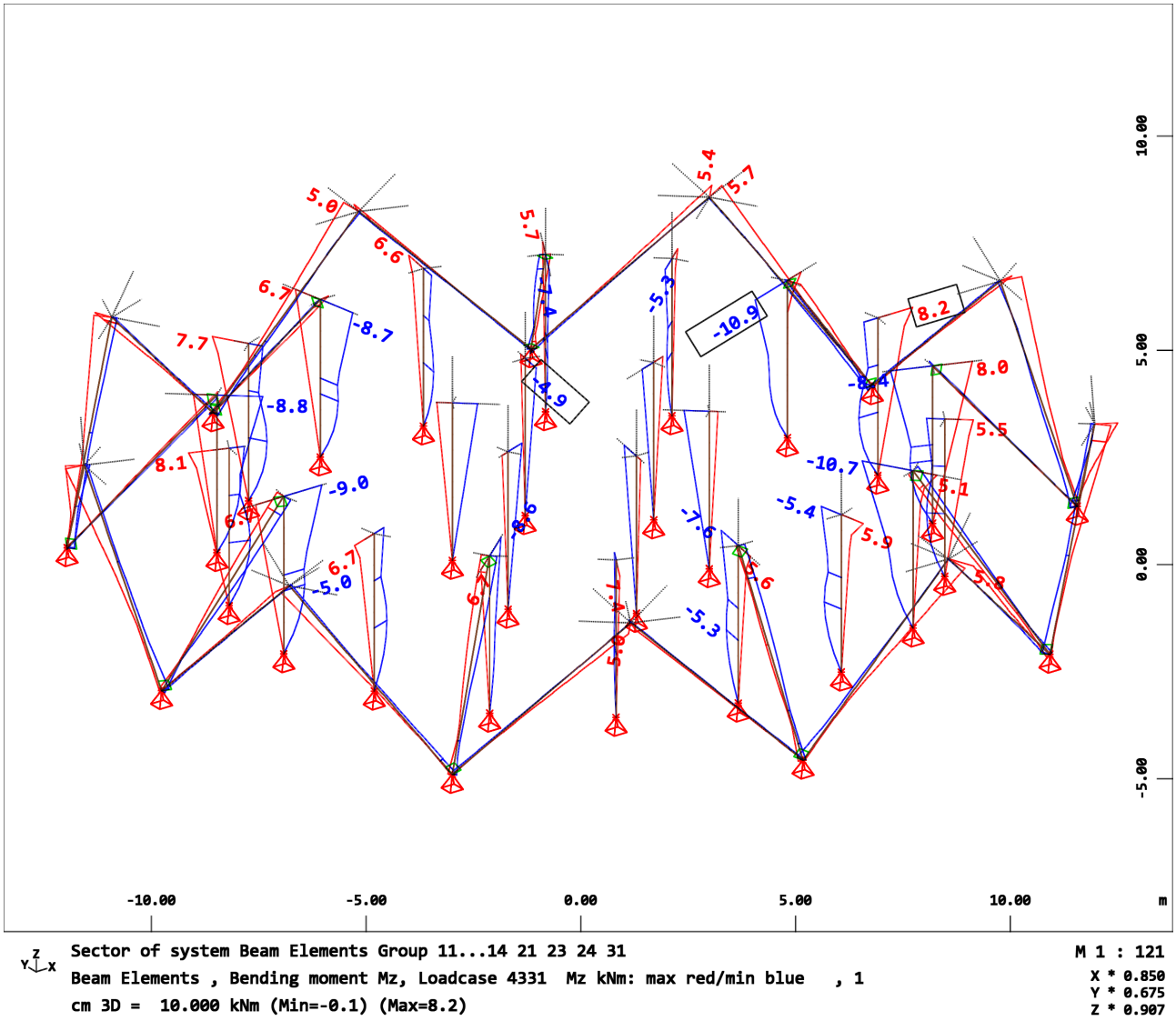
P12_V_G-L1_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers ground to Level 1 (groups 11, 12, 13, 14, 21, 23, 24, 31) N. Diagram scale 1 cm = 200 kN (on the 25.5 cm sheet); labels only where |value| > 239 kN. Envelope extremes: max 56.84, min -398.98 kN.



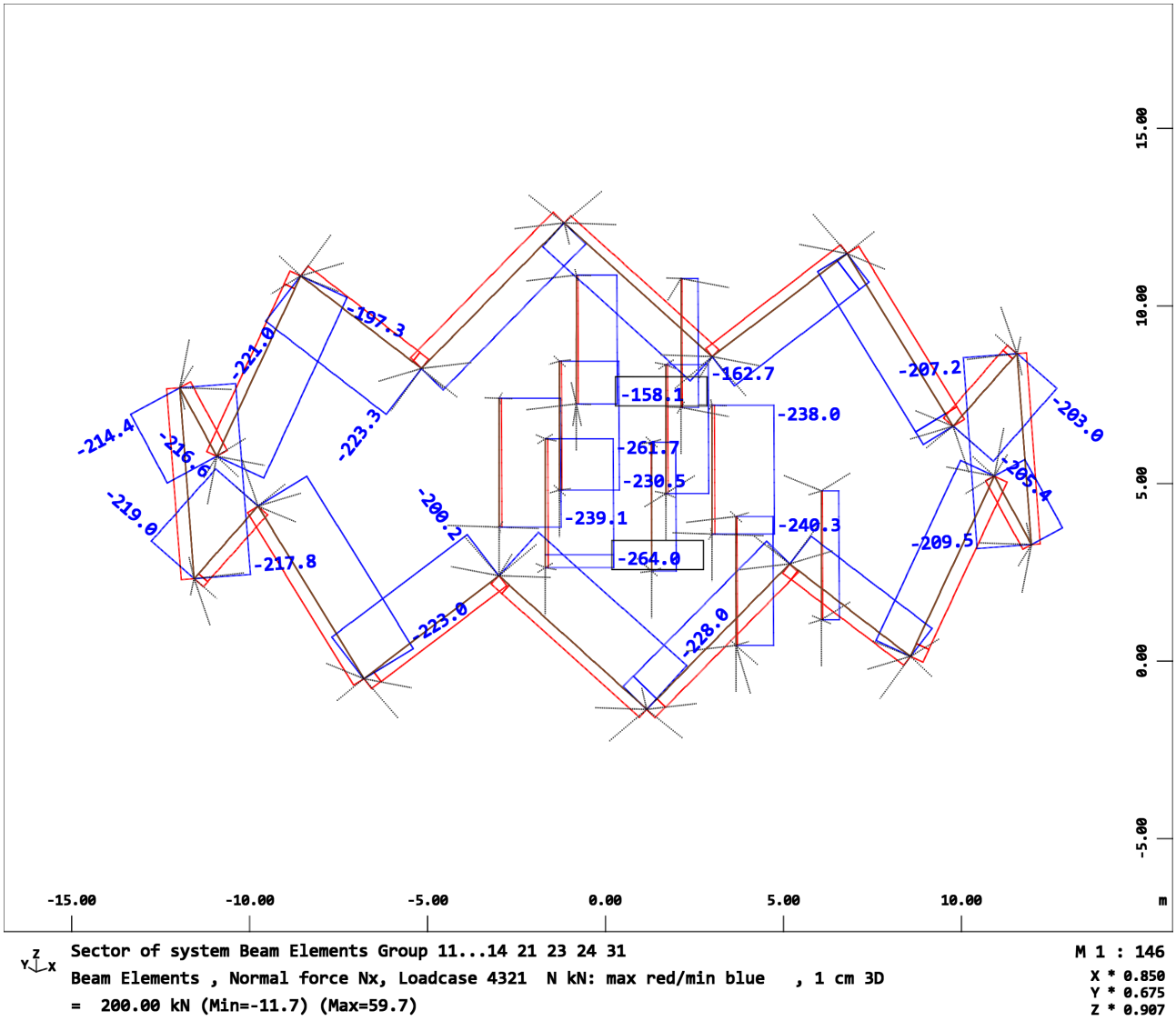
P13_V_G-L1_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers ground to Level 1 (groups 11, 12, 13, 14, 21, 23, 24, 31) My. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 3.2 kNm. Envelope extremes: max 7.08, min -7.08 kNm.



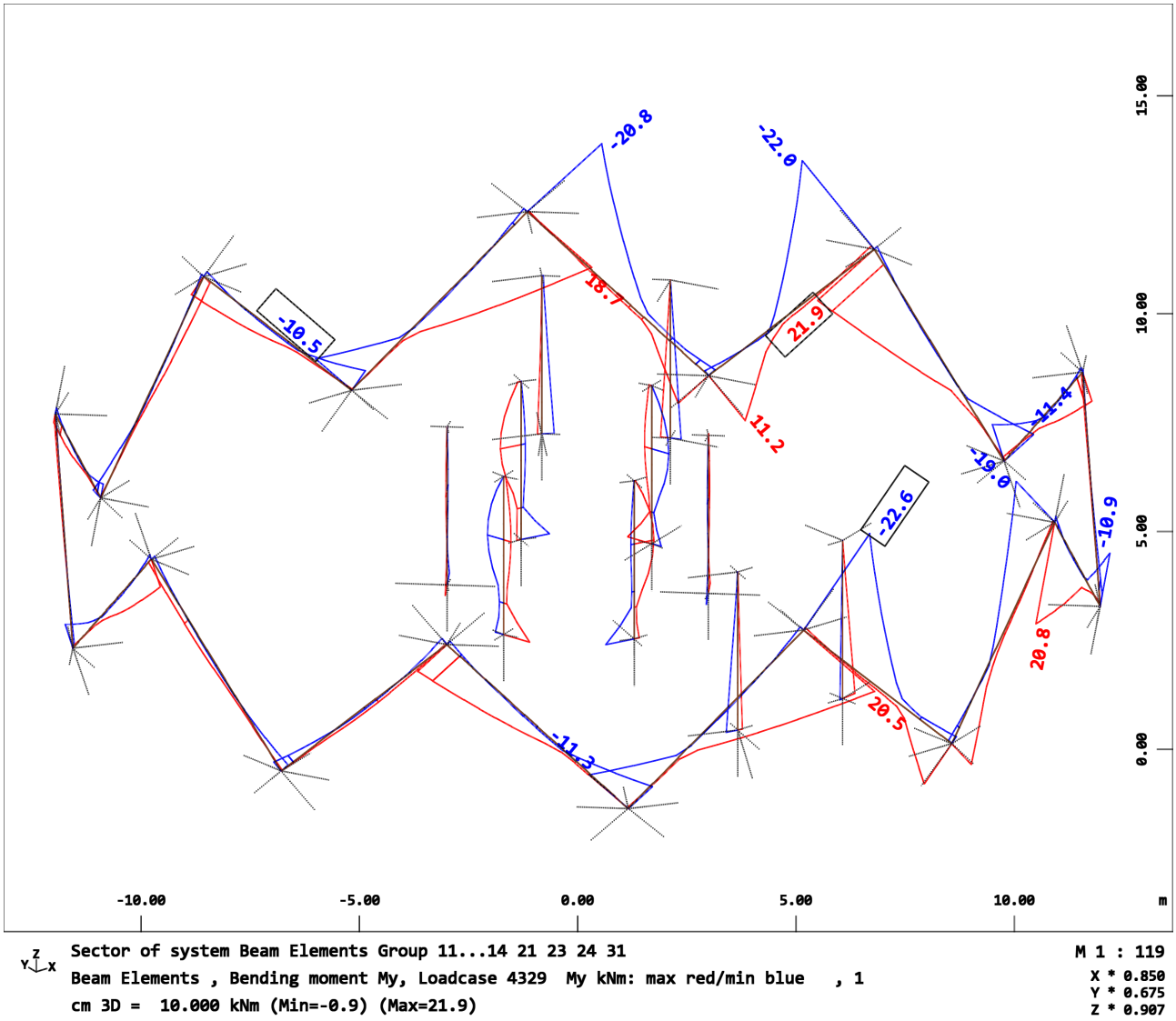
P14_V_G-L1_Mz

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers ground to Level 1 (groups 11, 12, 13, 14, 21, 23, 24, 31) Mz. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 4.9 kNm. Envelope extremes: max 8.2, min -10.93 kNm.



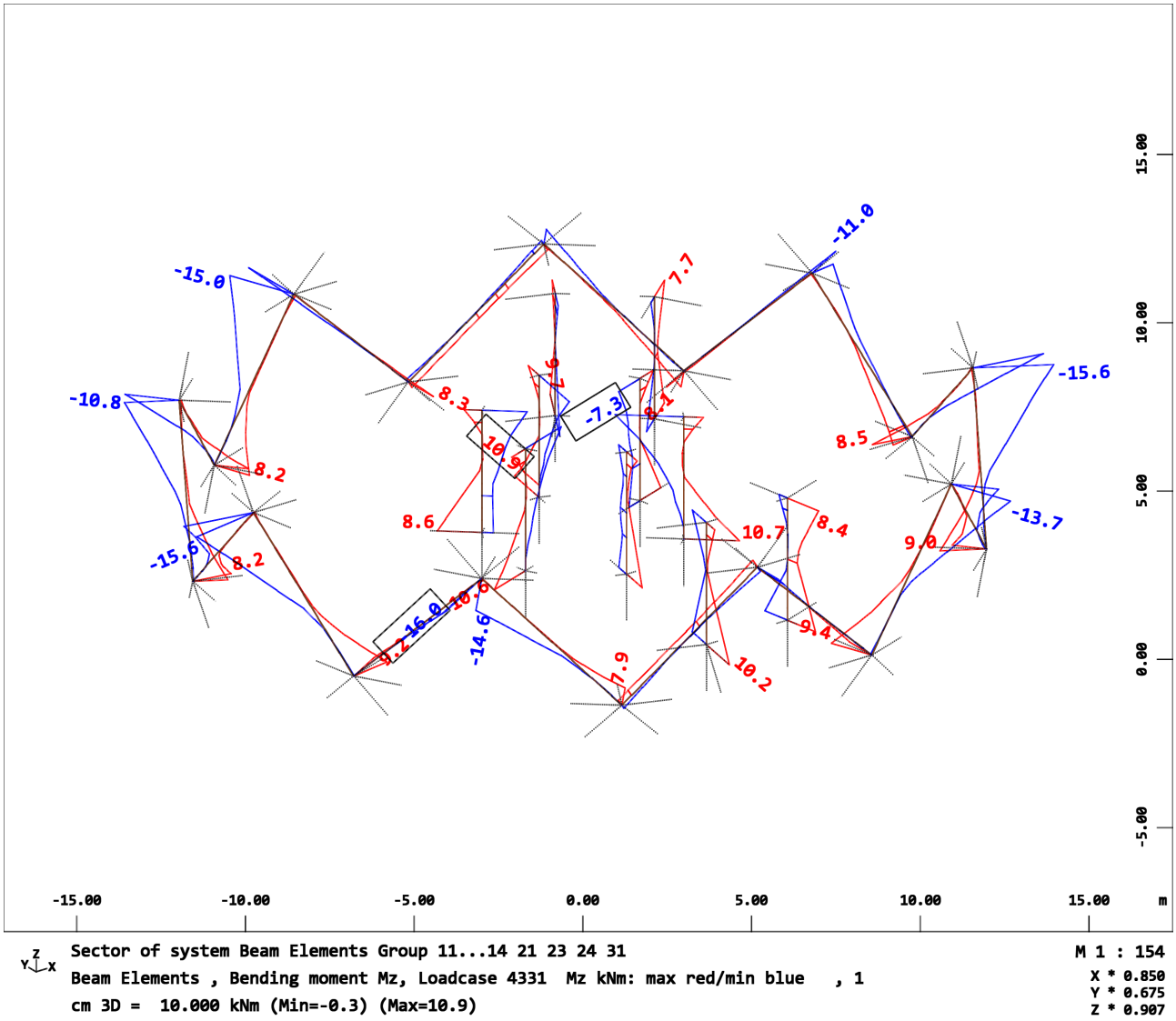
P15_V_L1-L2_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 1 to Level 2 (groups 11, 12, 13, 14, 21, 23, 24, 31) N. Diagram scale 1 cm = 200 kN (on the 25.5 cm sheet); labels only where |value| > 158 kN. Envelope extremes: max 59.71, min -264.01 kN.



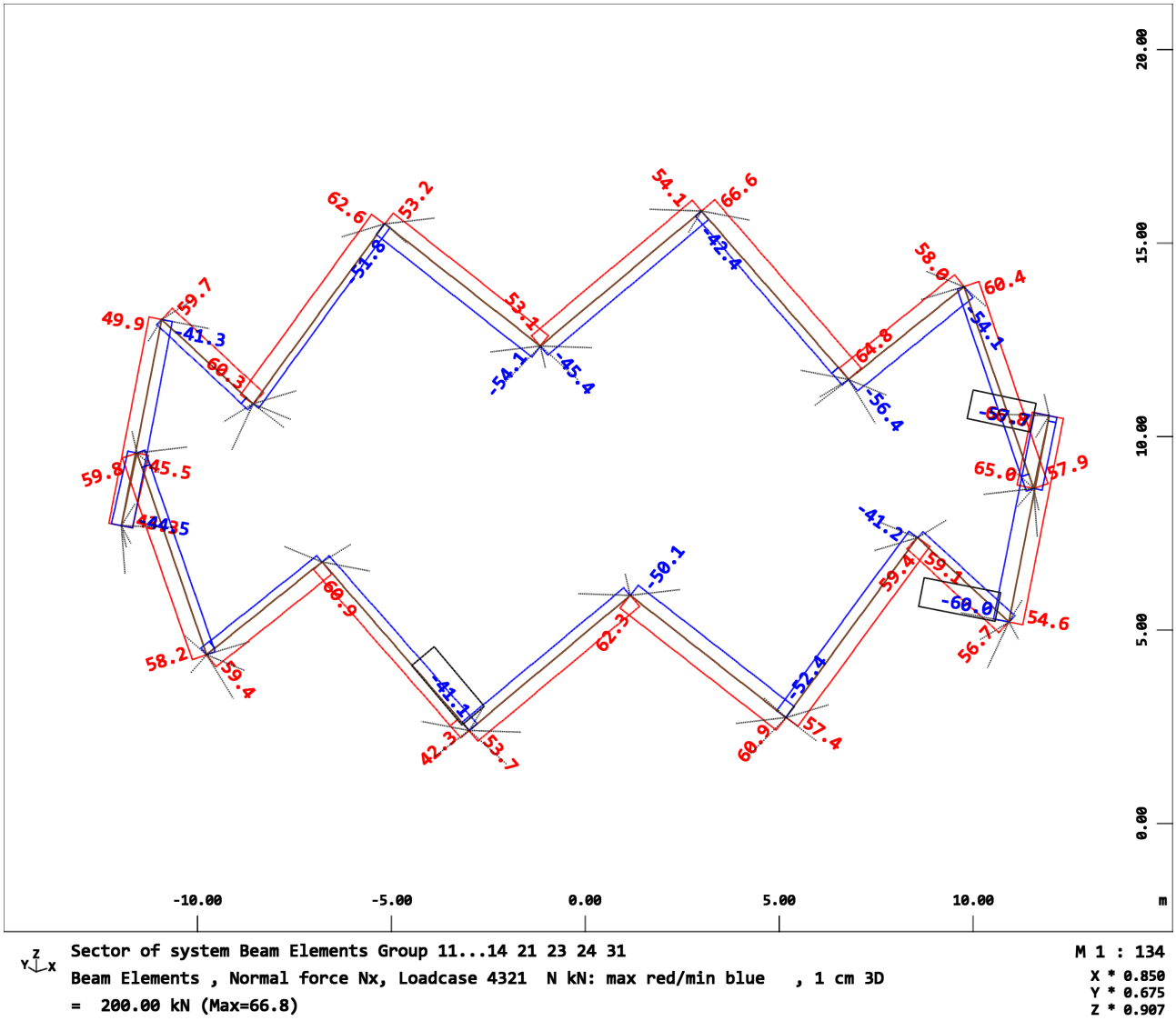
P16_V_L1-L2_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 1 to Level 2 (groups 11, 12, 13, 14, 21, 23, 24, 31) My. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 10 kNm. Envelope extremes: max 21.91, min -22.58 kNm.



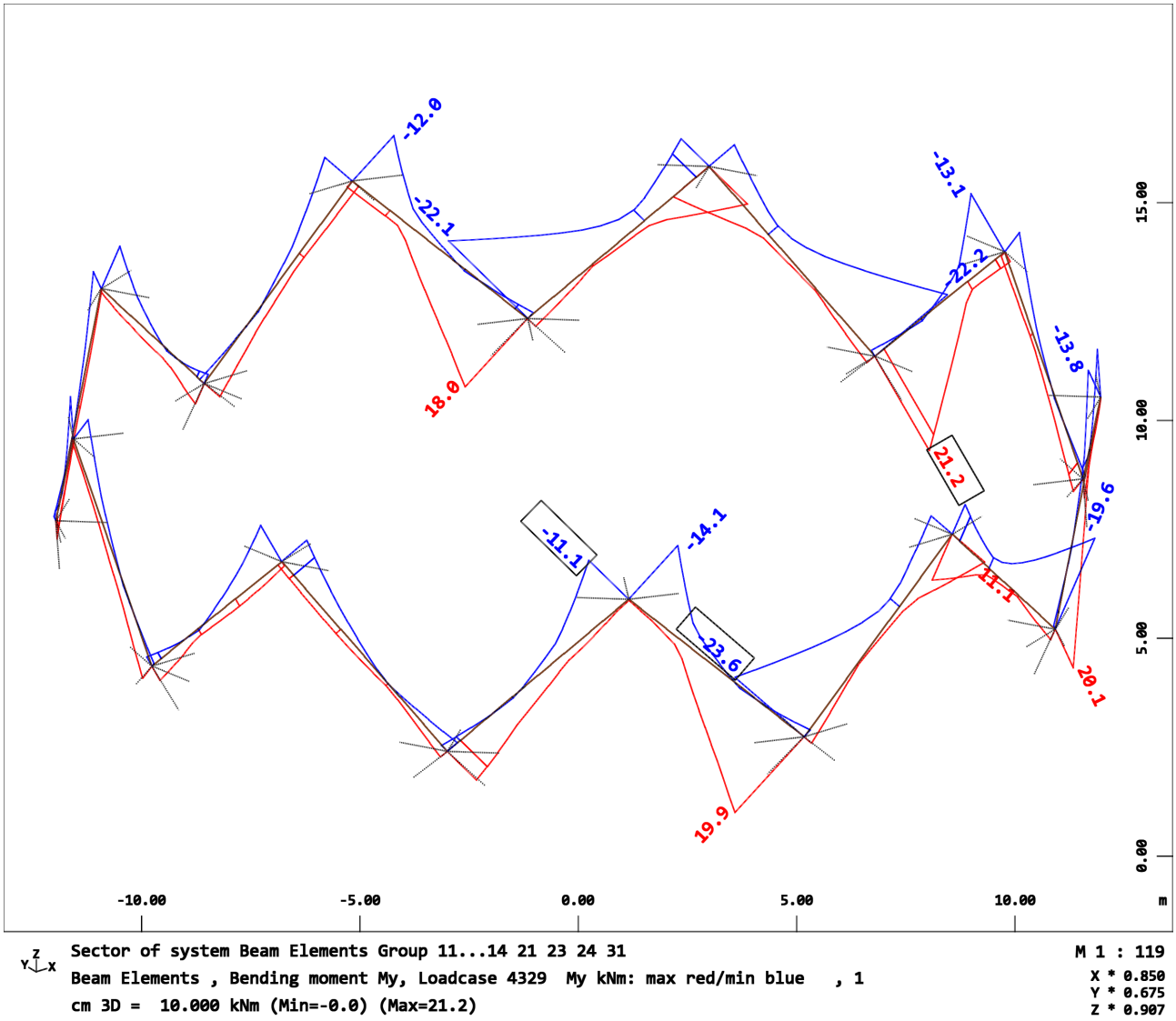
P17_V_L1-L2_Mz

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 1 to Level 2 (groups 11, 12, 13, 14, 21, 23, 24, 31) Mz. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 7.2 kNm. Envelope extremes: max 10.91, min -15.96 kNm.



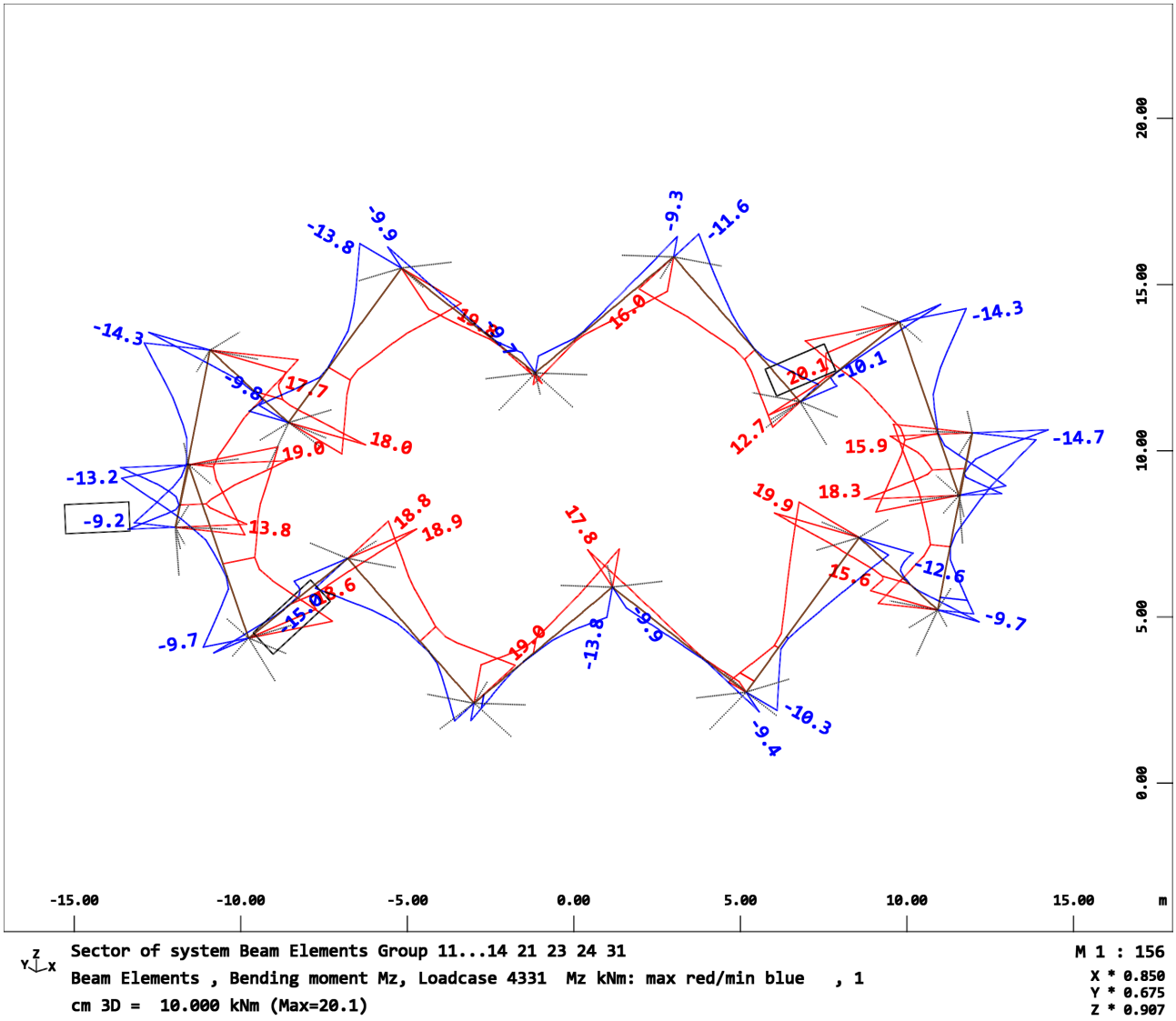
P18_V_L2-RF_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 2 to roof (groups 11, 12, 13, 14, 21, 23, 24, 31) N. Diagram scale 1 cm = 200 kN (on the 25.5 cm sheet); labels only where |value| > 40 kN. Envelope extremes: max 66.78, min -60 kN.



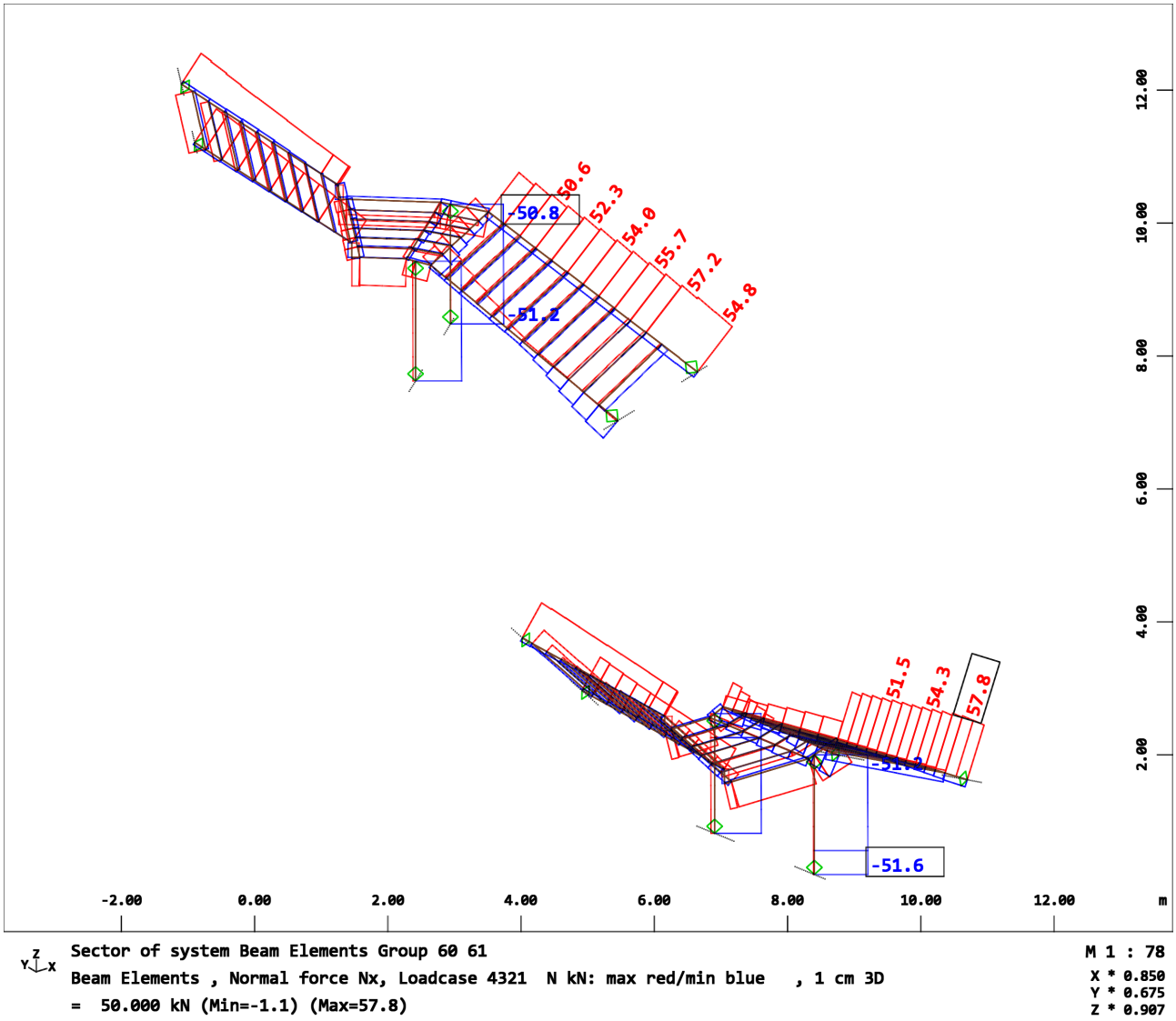
P19_V_L2-RF_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 2 to roof (groups 11, 12, 13, 14, 21, 23, 24, 31) My. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 11 kNm. Envelope extremes: max 21.21, min -23.63 kNm.



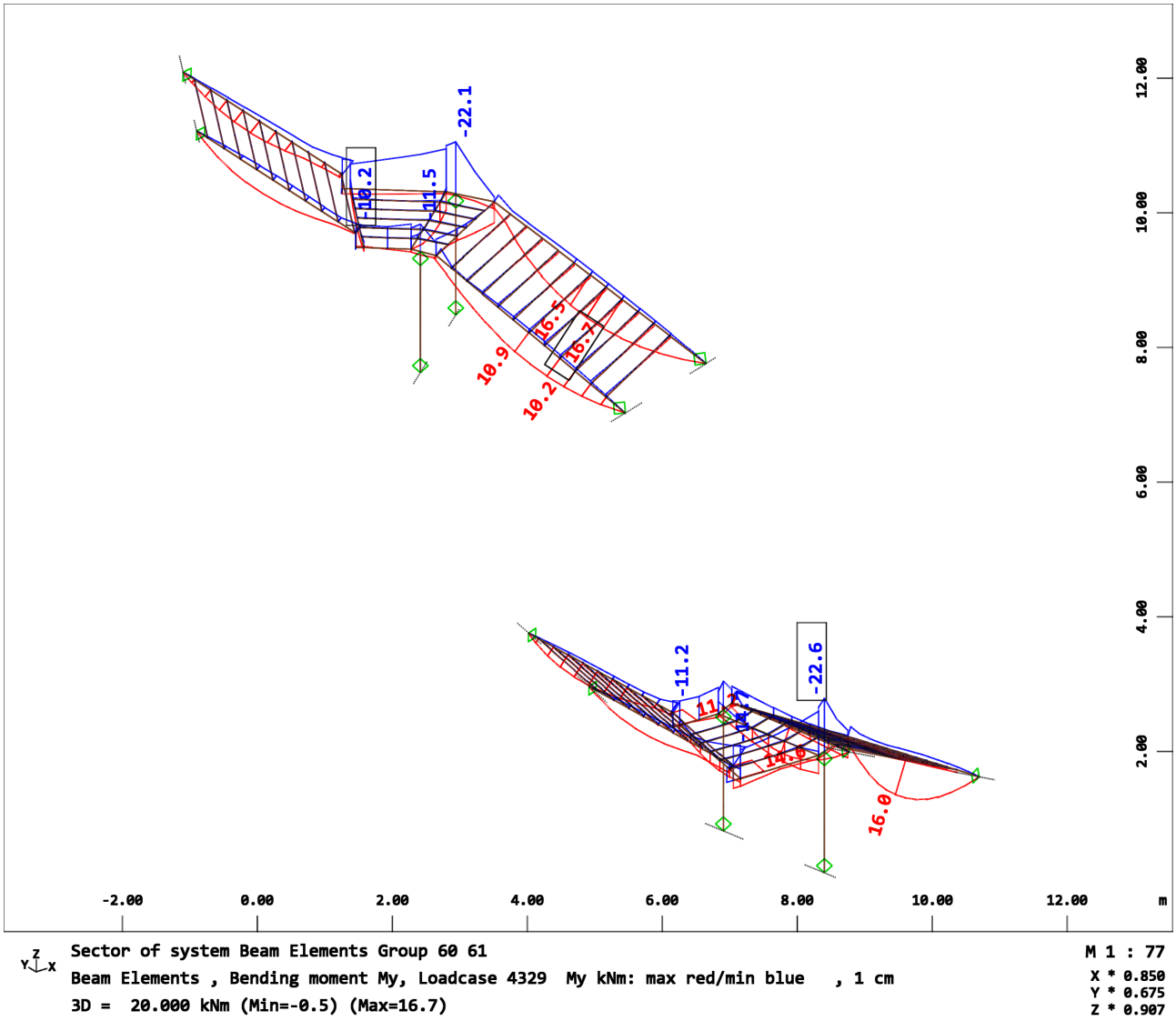
P20_V_L2-RF_Mz

Mz [kNm], ULS envelope: max LC 4331 (red) and min LC 4332 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. columns and rakers Level 2 to roof (groups 11, 12, 13, 14, 21, 23, 24, 31) Mz. Diagram scale 1 cm = 10 kNm (on the 25.5 cm sheet); labels only where |value| > 9 kNm. Envelope extremes: max 20.05, min -14.96 kNm.



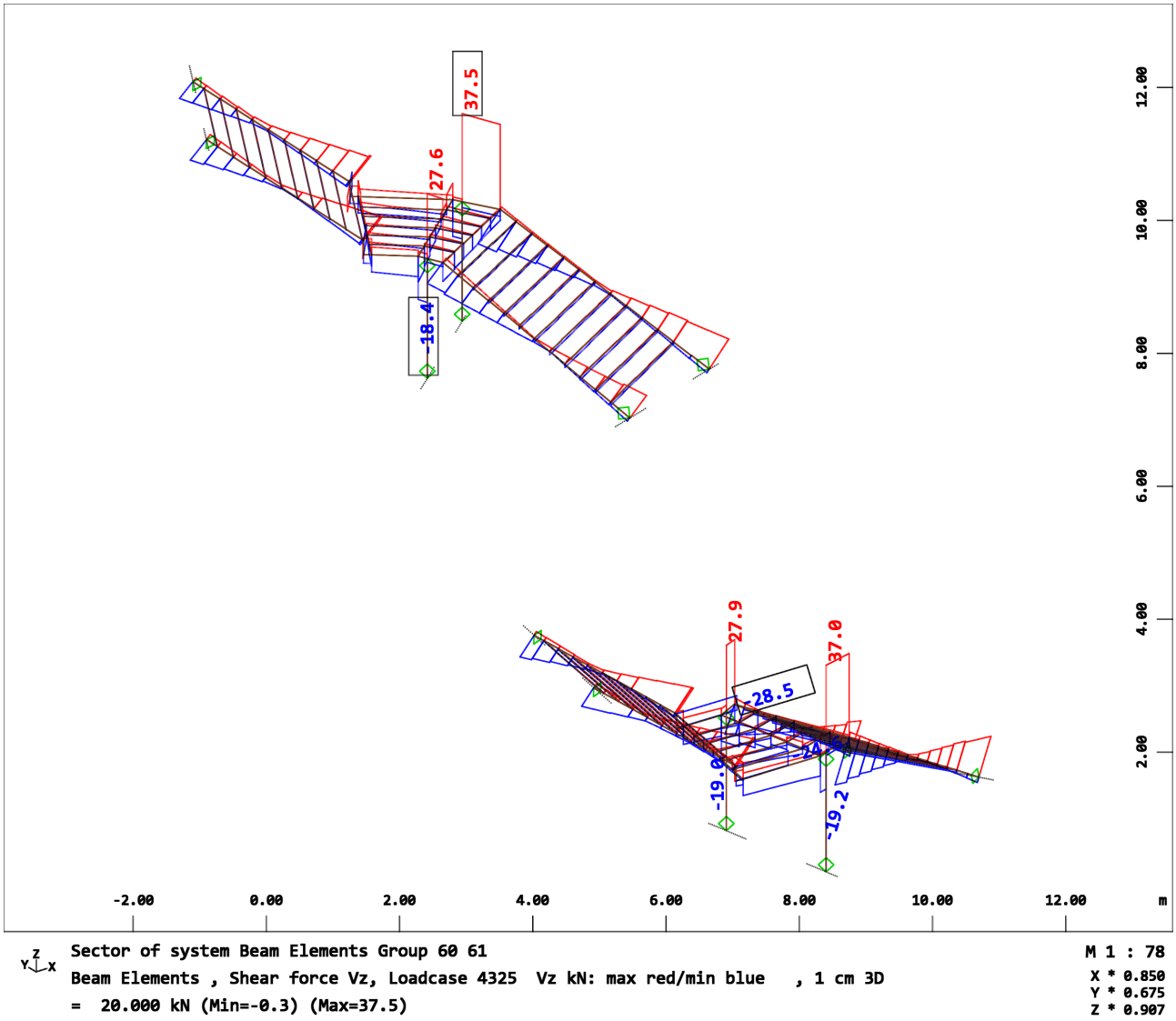
P21_S_int_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. internal stairs (groups 60, 61) N. Diagram scale 1 cm = 50 kN (on the 25.5 cm sheet); labels only where |value| > 49 kN. Envelope extremes: max 57.75, min -51.58 kN.



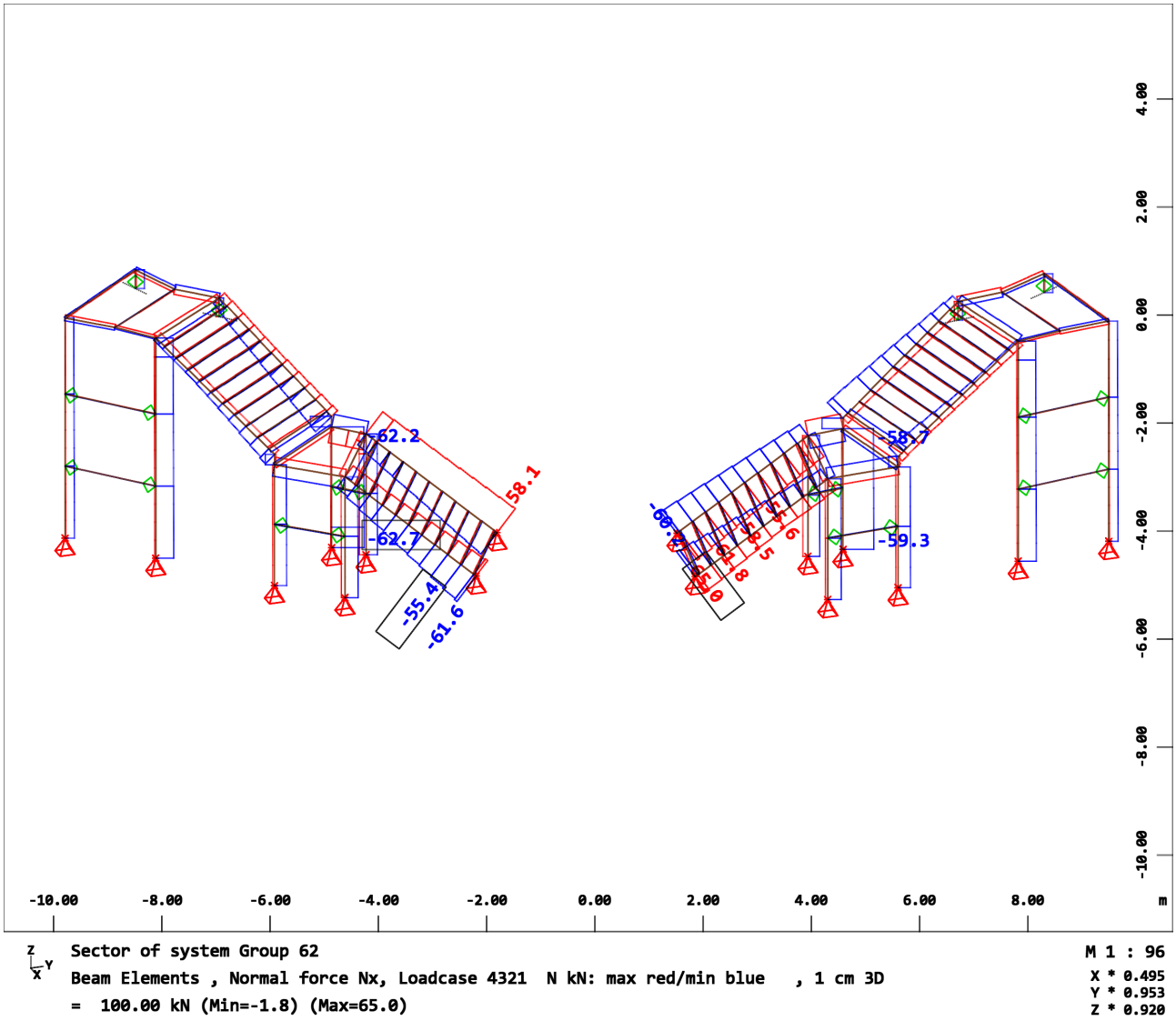
P22_S_int_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. internal stairs (groups 60, 61) My. Diagram scale 1 cm = 20 kNm (on the 25.5 cm sheet); labels only where |value| > 10 kNm. Envelope extremes: max 16.69, min -22.58 kNm.



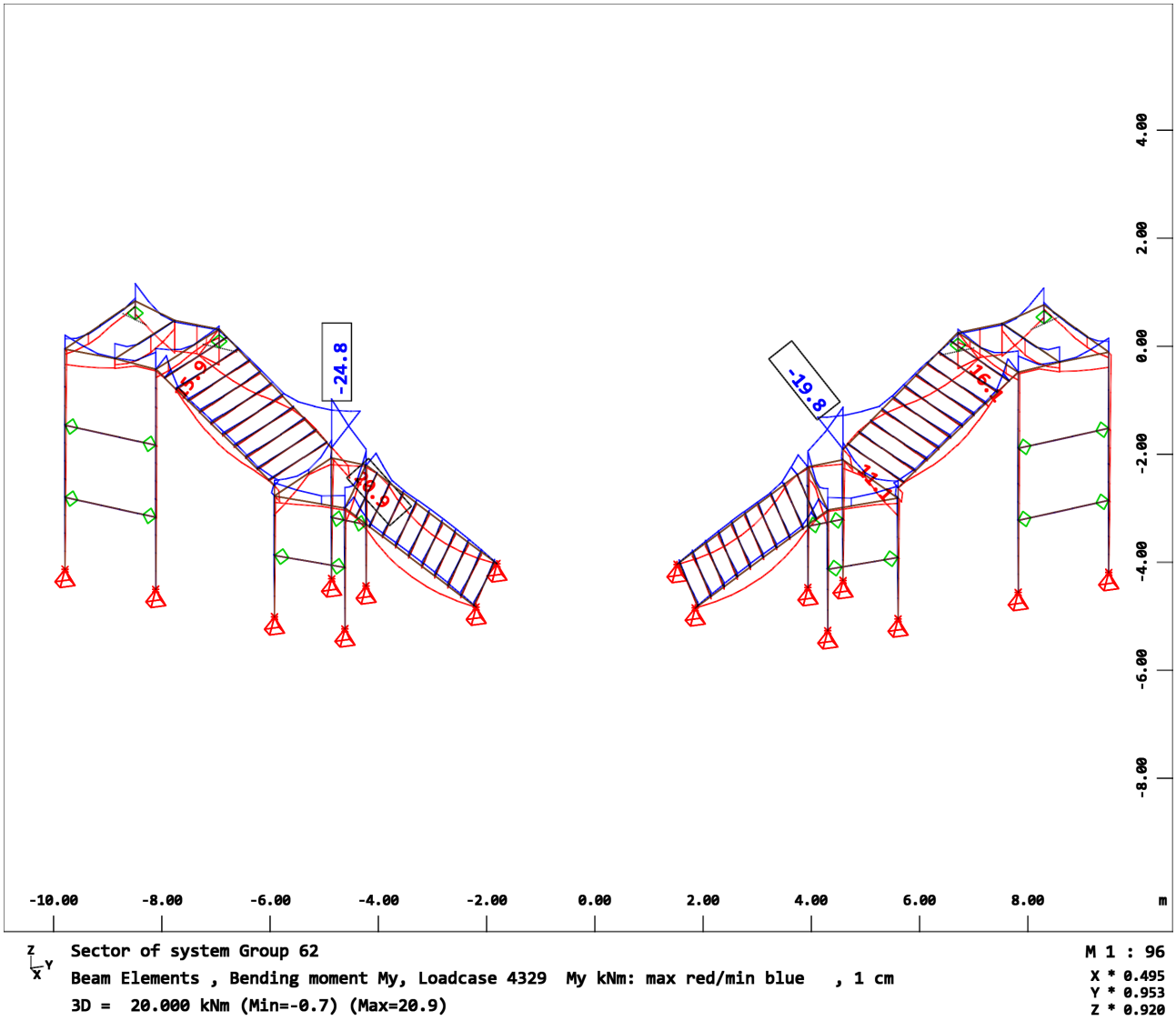
P23_S_int_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. internal stairs (groups 60, 61) Vz. Diagram scale 1 cm = 20 kN (on the 25.5 cm sheet); labels only where |value| > 17 kN. Envelope extremes: max 37.47, min -28.53 kN.



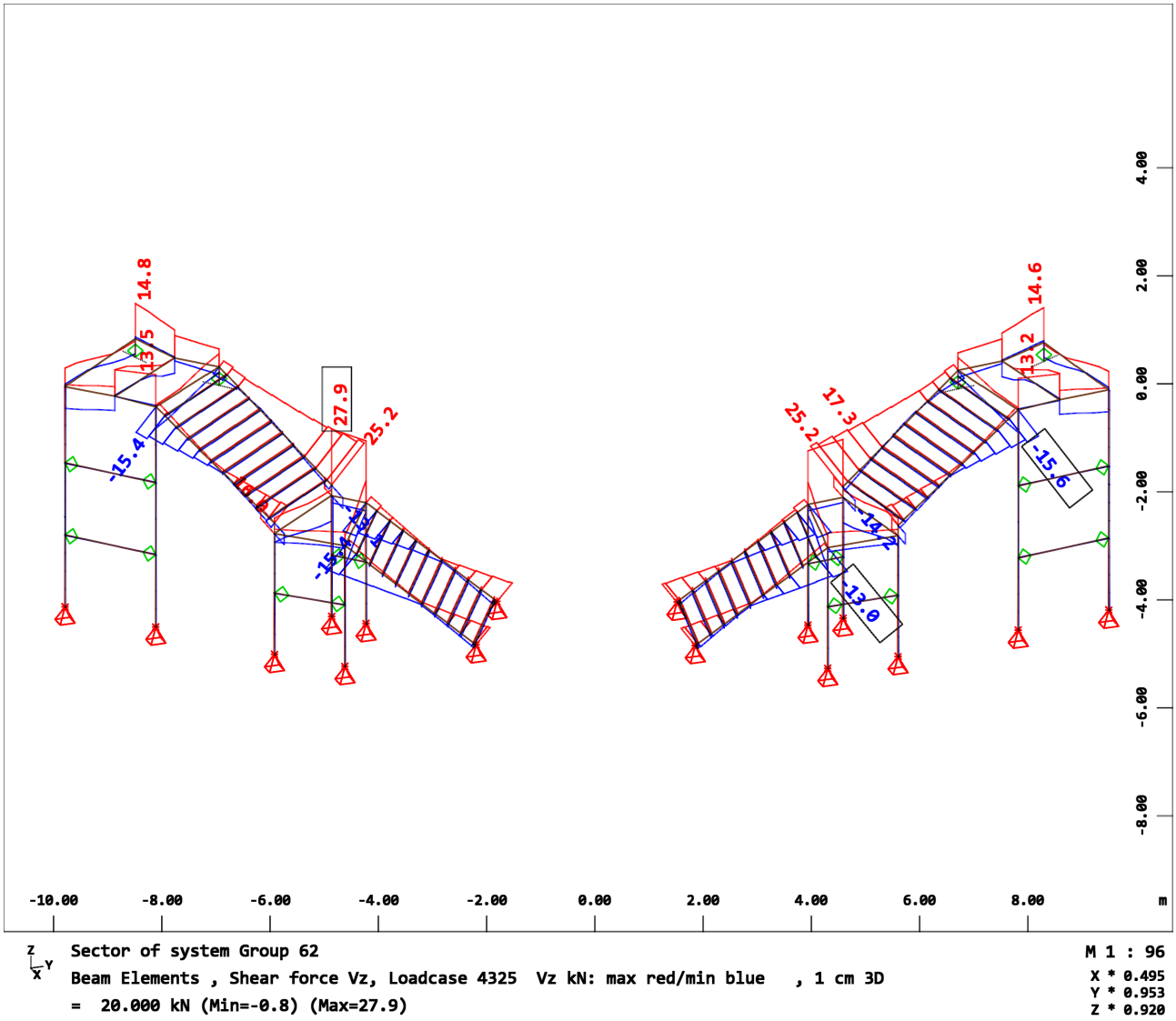
P24_S_ext_N

N [kN], ULS envelope: max LC 4321 (red) and min LC 4322 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. external stair (group 62) N. Diagram scale 1 cm = 100 kN (on the 25.5 cm sheet); labels only where |value| > 55 kN. Envelope extremes: max 64.98, min -62.71 kN.



P25_S_ext_My

My [kNm], ULS envelope: max LC 4329 (red) and min LC 4330 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. external stair (group 62) My. Diagram scale 1 cm = 20 kNm (on the 25.5 cm sheet); labels only where |value| > 11 kNm. Envelope extremes: max 20.93, min -24.77 kNm.



P26_S_ext_Vz

Vz [kN], ULS envelope: max LC 4325 (red) and min LC 4326 (blue) over the 43 ULS STR combinations 101-130, 135, 141-148, 151-154. external stair (group 62) Vz. Diagram scale 1 cm = 20 kN (on the 25.5 cm sheet); labels only where |value| > 13 kN. Envelope extremes: max 27.85, min -15.62 kN.